5G RAN

Network Slicing Feature Parameter Description

Issue Draft B

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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Modified the user-rate-based dynamic network slice resource management mechanism. For details, see: • **4.1.4.2.7 User-Rate-based Dynamic Network Slice Resource Management** • **4.2.2 Impacts** • **4.4.1.1 Data Preparation** • **4.4.1.2 Using MML Commands**	Added parameters: • **NRDUCellQciBearer**.***DynResUlTargetRate*** • **NRDUCellQciBearer**.***DynResDlTargetRate***	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

- Editorial changes

  Added the requirements for the RB control policy when the single-UE multi-slice RB management policy is enabled. For details, see **4.1.4.2.3 Single-UE Multi-Slice RB Management Policy**.

  Deleted the "Network Slice Group Configuration Consistency Query" section, and moved its content to **4.1.4.1.1 Configuring a Network Slice Group**. For details, see **4.1 Principles** and **4.1.4.1.1 Configuring a Network Slice Group**.

Revised descriptions of the hardware requirements and setting notes for user-rate-based dynamic network slice resource management. For details, see **4.3.3 Hardware** and **4.4.1.1 Data Preparation**.

Added descriptions of the impact relationships of "user-rate-based dynamic network slice resource management" with UL and DL Decoupling and independent SUL deployment. For details, see **4.3.2.3 Function Impacts**.

Added descriptions of the user-rate-based dynamic network slice resource management mechanism in an NR DU cell TRP. For details, see **4.1.4.2.7 User-Rate-based Dynamic Network Slice Resource Management**.

Added descriptions of the impact relationships of "user-rate-based dynamic network slice resource management" with the following functions: downlink intra-band CA, downlink intra-FR inter-band CA, uplink intra-band CA, and uplink intra-FR inter-band CA. For details, see **4.3.2.3 Function Impacts**.

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 04 (2025-08-30).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for network slicing in high frequency bands. For details, see: • **2 About This Document** • **4 Network Slicing Introduction**	Modified parameters: Added support for high-frequency scenarios by the following parameters: • **NRDUCellRes**.***CellResId*** • **NRDUCellRes**.***ResUsageObject*** • **ADMISSION_BASED_ON_NS_ID_SW** option of the **NRCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter • **ENHANCED_NETWORKSLICE_SW** option of the **NRCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter • **RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter	High-frequency TDD	5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Optimized the method for observing network slicing. For details, see **4.4.2 Activation Verification**.	None	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added user-rate-based dynamic network slice resource management. For details, see: • **2.3 Differences** • **4.1.4.2 Enhanced Resource Management** • **4.1.4.2.7 User-Rate-based Dynamic Network Slice Resource Management** • **4.2.2 Impacts** • **4.3.2.1 Prerequisite Functions** • **4.3.2.2 Mutually Exclusive Functions** • **4.3.2.3 Function Impacts** • **4.4.1.1 Data Preparation** • **4.4.1.2 Using MML Commands**	Modified parameter: Added the **DYNAMIC_NETWORKSLICE_SW** option to the **NRDUCellCommonSw**.***ServiceDiffAlgoExtSwitch*** parameter.	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added a mutually exclusive relationship between "user-rate-based dynamic QCI resource management" and "NR DU cell resource management between network slices". For details, see **4.3.2.2 Mutually Exclusive Functions**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added an impact relationship between uplink 4-layer transmission for a single FWA user and uplink scheduling on a per RB basis. For details, see **4.3.2.3 Function Impacts**.	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations)
Added a mutually exclusive relationship between FSD transmission mode and "NR DU cell resource management between network slices". For details, see **4.3.2.2 Mutually Exclusive Functions**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship between "enhanced L4S congestion control based on rate adjustment" and "NR DU cell resource management between network slices". For details, see **4.3.2.3 Function Impacts**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added an impact relationship between "NR DU cell resource management between network slices" and "flexible experience differentiation". For details, see **4.3.2.3 Function Impacts**.	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes

  Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed. Examples of such changes include changes in the names of reference documents.

2 About This Document

2.1 General Statements

Purpose

This document is intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-050201	Network Slicing Introduction	**4 Network Slicing Introduction**

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Network Slicing Introduction	Supported only in SA networking	**4 Network Slicing Introduction**

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Network Slicing Introduction	Supported in both high and low frequency bands, with the following differences: ● Resource sharing between network slice groups and RFSP groups is supported only in low frequency bands. For details, see **4.1.3 NR Cell Resource Management Between Network Slices** and **4.1.4.2.4 Resource Sharing Between Network Slice Groups and RFSP Groups**. ● The TTI-based control policy is supported only in low frequency bands. For details, see **TTI-based Control Policy** in **4.1.4.1.5 Specifying an RB Resource Control Policy**. ● Downlink type 1 scheduling is supported only in low frequency bands. For details, see **4.1.4.2.1 Downlink Type 1 Scheduling**. ● Uplink scheduling on a per RB basis is supported only in low frequency bands. For details, see **4.1.4.2.2 Uplink Scheduling on a per RB Basis**. ● The single-UE multi-slice RB management policy is supported only in low frequency bands. For details, see **4.1.4.2.3 Single-UE**	**4 Network Slicing Introduction**

Function Name	Difference	Chapter/Section
	Multi-Slice RB Management Policy. • Operator-specific resource sharing enhancement is supported only in low frequency bands. For details, see **4.1.4.2.5 Operator-specific Resource Sharing Enhancement**. • Interference randomization is supported only in low frequency bands. For details, see **4.1.4.2.6 Interference Randomization**. • User-rate-based dynamic network slice resource management is supported only in low frequency bands. For details, see **4.1.4.2.7 User-Rate-based Dynamic Network Slice Resource Management**.	

3 Overview

Background of Network Slicing

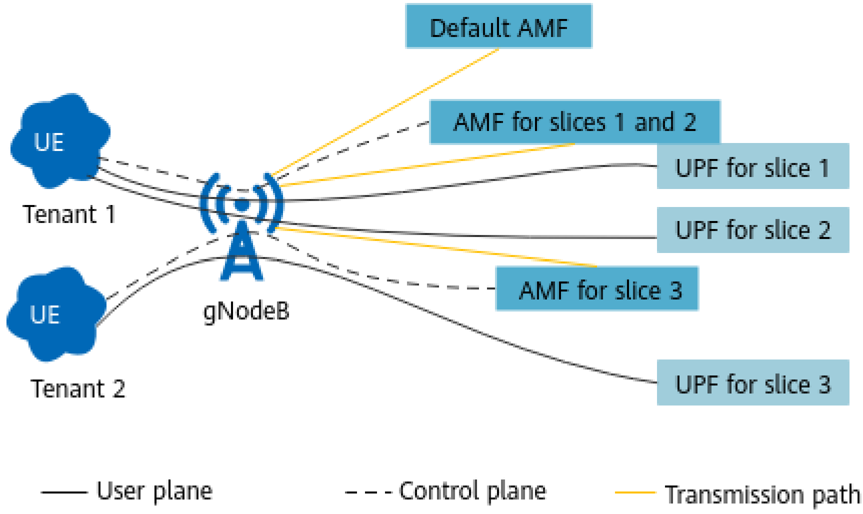
5G enables the integration of the communications industry with vertical industries. The ultimate goal of 5G is to digitize all industries. To achieve this goal, 5G networks will be designed to meet three key requirements: independent service operation, secure isolation, and Service Level Agreement (SLA) assurance.

Traditional networks cannot provide independent operation and secure isolation tailored to different services and lack efficient SLA assurance for all services. To resolve these shortcomings, the network slicing function (defined in section 16.3 "Network Slicing" in 3GPP TS 38.300 V15.6.0) is introduced.

Network slicing allows an operator's physical network to be sliced into multiple E2E virtual logical networks. The sliced networks are isolated from each other and provide customizable network capabilities, including latency, bandwidth, number of connections, and reliability, for industry customers. In this way, multiple logical networks can run on an operator's physical network to provide tailored services.

5G network slicing works under SA networking and requires support from multiple domains, including terminals, a radio access network (RAN), a transport network, and a core network (CN). **Figure 3-1** illustrates the network slice interconnection model.

Figure 3-1 Network slice interconnection model



Network Slice Identifier

A network slice is uniquely identified by an E2E identifier, Single Network Slice Selection Assistance Information (S-NSSAI), as defined in section 5.15.2 "Identification and selection of a Network Slice: the S-NSSAI and the NSSAI" in 3GPP TS 23.501 V15.5.0. An S-NSSAI contains:

- Slice/service type (SST): This is a mandatory field that identifies the type of slice. It consists of eight bits and can be set to a value ranging from 0 to 255. Values ranging from 0 to 127 are standardized SST values that can be applied to the entire network. **Table 3-1** lists six standardized SST values defined in the current protocol release. However, values ranging from 128 to 255 are customized SST values that are valid only in an operator-specific network.

Table 3-1 SST values defined in 3GPP specifications

Type	Value	Description
eMBB	1	Suitable for 5G enhanced Mobile Broadband (eMBB).
URLLC	2	Suitable for ultra-reliable low-latency communication (URLLC).
MIoT	3	Suitable for Massive Internet of Things (MIoT).
V2X	4	Suitable for vehicle-to-everything (V2X) services.
HMTC	5	Suitable for high-performance machine-type communication.
HDLLC	6	Suitable for high-data-rate and low-latency communication.

- Slice Differentiator (SD): This is an optional field used to differentiate between slices that share the same SST value. It consists of 24 bits and can be customized by operators.

NOTE

For ease of description, S-NSSAI is used in this document to refer to a network slice identifier.

To have a quick look at network slicing, see **(Video) What Is Network Slicing**.

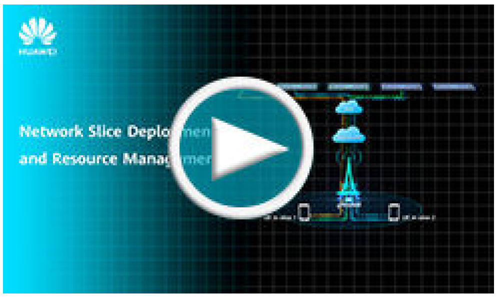
4 Network Slicing Introduction

4.1 Principles

The RAN supports network slice deployment so that it can be aware of network slices configured on the CN for E2E connection required in network slicing. In addition, the RAN provides radio resource management and admission control for network slices to meet the SLA requirements of different services and ensure user experience. To achieve this, the Network Slicing Introduction package offers the following functions:

- **Deployment and awareness**: Once network slice deployment is complete, the gNodeB becomes aware of the network slices on the CN, and E2E identification of network slices is enabled. For details, see **4.1.1 Network Slice Deployment and Awareness**.
- **Admission control**: The gNodeB performs admission control to achieve resource isolation based on the network slices used for UEs and the maximum number of RRC_CONNECTED UEs allowed in the corresponding network slice group. For details, see **4.1.2 Network Slice Admission Control**.
- **NR cell resource management between network slices**: NR cell resource management controls the number of RRC_CONNECTED UEs. This function manages UE access and handovers by associating S-NSSAIs with network slice groups and configuring the maximum number of RRC_CONNECTED UEs for these groups. For details, see **4.1.3 NR Cell Resource Management Between Network Slices**.
- **NR DU cell resource management between network slices**: NR DU cell resource management controls the RBs available to a cell. The gNodeB allocates resources to different network slices based on the resource allocation policy to ensure differentiated user experience. For details, see **4.1.4 NR DU Cell Resource Management Between Network Slices**.

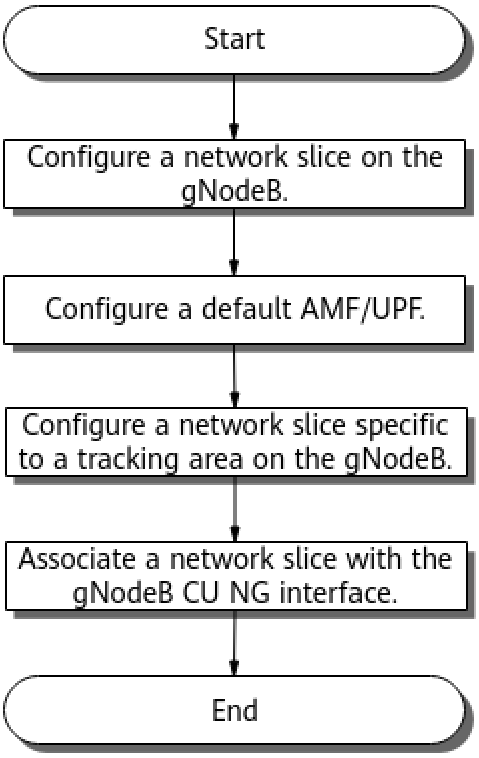
To have a quick look at network slice deployment and resource management, see **(Video) Network Slice Deployment and Resource Management**.



4.1.1 Network Slice Deployment and Awareness

A network slice deployment process includes the following sequential operations: network slice configuration on the gNodeB; configuration of the default AMF and UPF; network slice configuration specific to a tracking area; association of network slices with the NG interface. These operations ensure normal access and service running for network slice UEs. Note that network slices are also configured on the CN in granularity of tracking areas. (AMF is short for access and mobility management function. UPF is short for user plane function.) **Figure 4-1** shows this process. Once network slice deployment is complete, the gNodeB becomes aware of the network slices on the CN.

Figure 4-1 Network slice deployment process



Configuring a Network Slice on the gNodeB

The **gNBNetworkSlice** MO is used to configure a network slice on the gNodeB. For details, see **4.4.1.1 Data Preparation**.

The **gNBNetworkSlice.*SliceServiceType*** and **gNBNetworkSlice.*SliceDifferentiator*** parameters collectively determine a network slice identifier, or S-NSSAI. Network slices with the same S-NSSAI but different **gNBNetworkSlice.*OperatorId*** values are treated as different network slices.

Configuring a Default AMF/UPF

Network slice deployment requires the configuration of a default AMF and a default UPF. A network slice UE can access a default AMF even when the RRCSetupComplete message sent by the UE contains neither the S-NSSAI nor the Temp ID, or contains both of them but neither is available. (Temp ID is also known as temporary mobile subscriber identity or TMSI.) A default UPF ensures that network slice UEs can properly use services. For details about the AMF selection mechanism, see **Table 4-1**.

Table 4-1 AMF selection

<table>
<tr><th>Scenario</th><th>S-NSSAI</th><th>Temp ID</th><th>AMF Selection</th><th>Remarks</th></tr>
<tr><td>1</td><td>√</td><td>x</td><td>The gNodeB selects the AMF specific to the NG interface associated with the network slice.</td><td rowspan="2">All network slice UEs can access the corresponding AMFs.</td></tr>
<tr><td>2</td><td>-</td><td>√</td><td>The gNodeB selects the AMF associated with the Temp ID.</td></tr>
<tr><td>3</td><td>x</td><td>x</td><td>The gNodeB selects a default AMF associated with an NG interface configured on the network.</td><td>A default AMF[a] is required to ensure that a network slice UE can access the default AMF when the UE indicates neither the S-NSSAI supported by the gNodeB nor the Temp ID.</td></tr>
<tr><td colspan="5">a: Default AMFs are configured at the operator level. In multi-operator scenarios, each operator requires at least one default AMF. The gNBCUNg.gNBCuNgId and gNBCUNg.CuNgType parameters collectively determine a default AMF. In addition, to ensure that network slice UEs properly use services, you need to configure a default UPF using the gNBCUNg.gNBCuNgId and gNBCUNg.CuNgType parameters.
Notes:
● The symbol "-" in the table indicates that the corresponding factor does not need to be considered.
● The symbol "√" in the table indicates that the item is carried and available.
● The symbol "x" in the table indicates that the item is not carried or available.</td></tr>
</table>

Configuring a Network Slice Specific to a Tracking Area on the gNodeB

On the CN, network slices are configured specific to tracking areas. As such, network slices on the gNodeB also need to be tracking-area-specific. A network slice specific to a tracking area is configured on the gNodeB by using the **gNBTaNs** MO. This way, a network slice on the gNodeB is associated with a tracking area. For details, see **4.4.1.1 Data Preparation**.

After a network slice on the gNodeB is associated with a tracking area, the gNodeB associates the network slice with the cells in the tracking area based on the mapping between the tracking area and the cells. The MML command **DSP NRCELLNS** or **DSP NRDUCELLNS** can be used to query the network slice information of a cell.

Associating a Network Slice with the gNodeB CU NG Interface

The **gNBCUNgNs** MO is used to configure a network slice specific to the gNodeB CU NG interface; that is, a network slice on the gNodeB is associated with an NG interface and the corresponding control-plane and user-plane endpoint groups. For details, see **4.4.1.1 Data Preparation**.

After a network slice on the gNodeB is associated with an NG interface and the control-plane or user-plane endpoint groups for the NG interface, the gNodeB can route data specific to the network slice to the target AMF or UPF. In addition, data specific to the network slice is isolated from the other data transmitted on the transport network due to the transport network VLAN ID in the NG interface configurations. For details about how to configure the NG interface, see the MML command examples for "NG Self-Setup" in *NG and Xn Self-Management*.

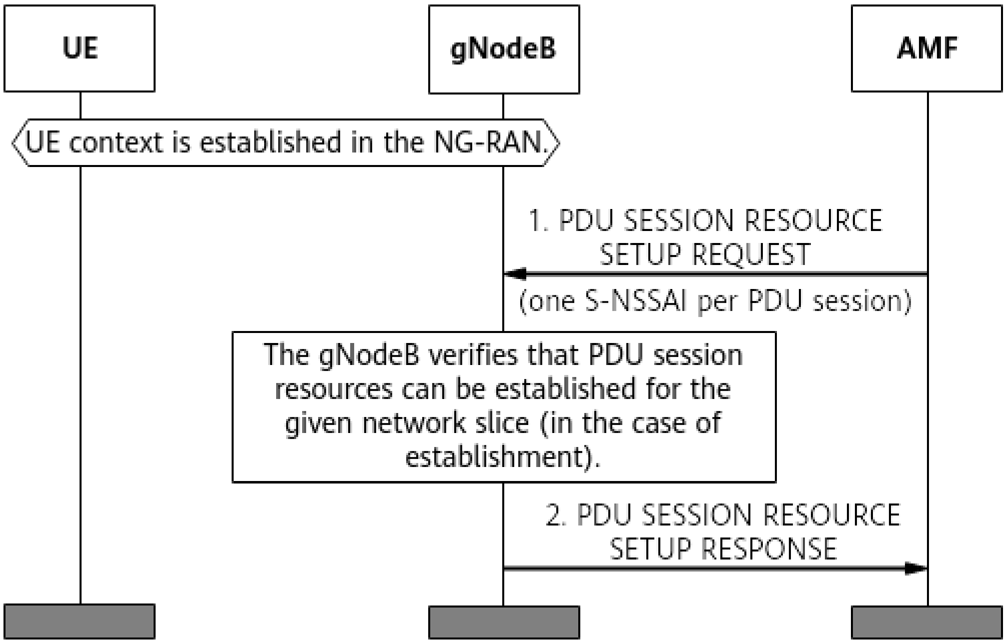
NOTICE

Each network slice must be associated with the control-plane endpoint group and user-plane endpoint group (configured using the **gNBCUNg** MO) corresponding to the NG interface. Otherwise, network slice services fail to be set up.

Network Slice Awareness

Figure 4-2 shows how the gNodeB becomes aware of network slices configured on the CN after network slice deployment is complete.

Figure 4-2 Network slice awareness in the gNodeB



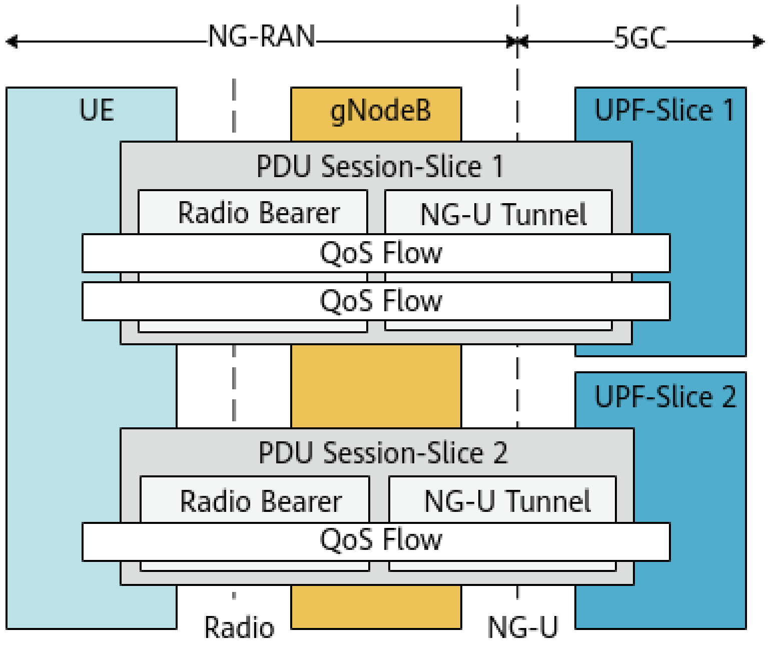
A UE sends a non-access stratum (NAS) message that contains S-NSSAI information to the gNodeB, which then forwards the NAS message to the CN. The CN then checks whether it supports the network slices reported from the UE. If it does not, the CN responds with information about supported network slices. If it does:

1. The CN sends a PDU SESSION RESOURCE SETUP REQUEST message that contains an S-NSSAI configured on the CN to the gNodeB to request that a PDU session be established. Note that each PDU session corresponds to one S-NSSAI.
2. The gNodeB then checks whether it supports the network slice identified by the received S-NSSAI. If it does, the gNodeB responds to the CN with a PDU SESSION RESOURCE SETUP RESPONSE message indicating that a PDU session has been successfully established. If it does not, the gNodeB responds to the CN with a PDU SESSION RESOURCE SETUP RESPONSE message indicating that a PDU session has failed to be established.

For details about messages and IEs involved in PDU session establishment, see section 9.2.1.1 "PDU SESSION RESOURCE SETUP REQUEST" in 3GPP TS 38.413 V15.4.0.

When the gNodeB becomes aware of the S-NSSAI corresponding to the PDU session, it obtains network slice and QoS information of each radio bearer based on radio bearer and QoS flow information in the PDU session. After that, the gNodeB performs differentiated processing specific to radio bearers. **Figure 4-3** shows this process in further detail.

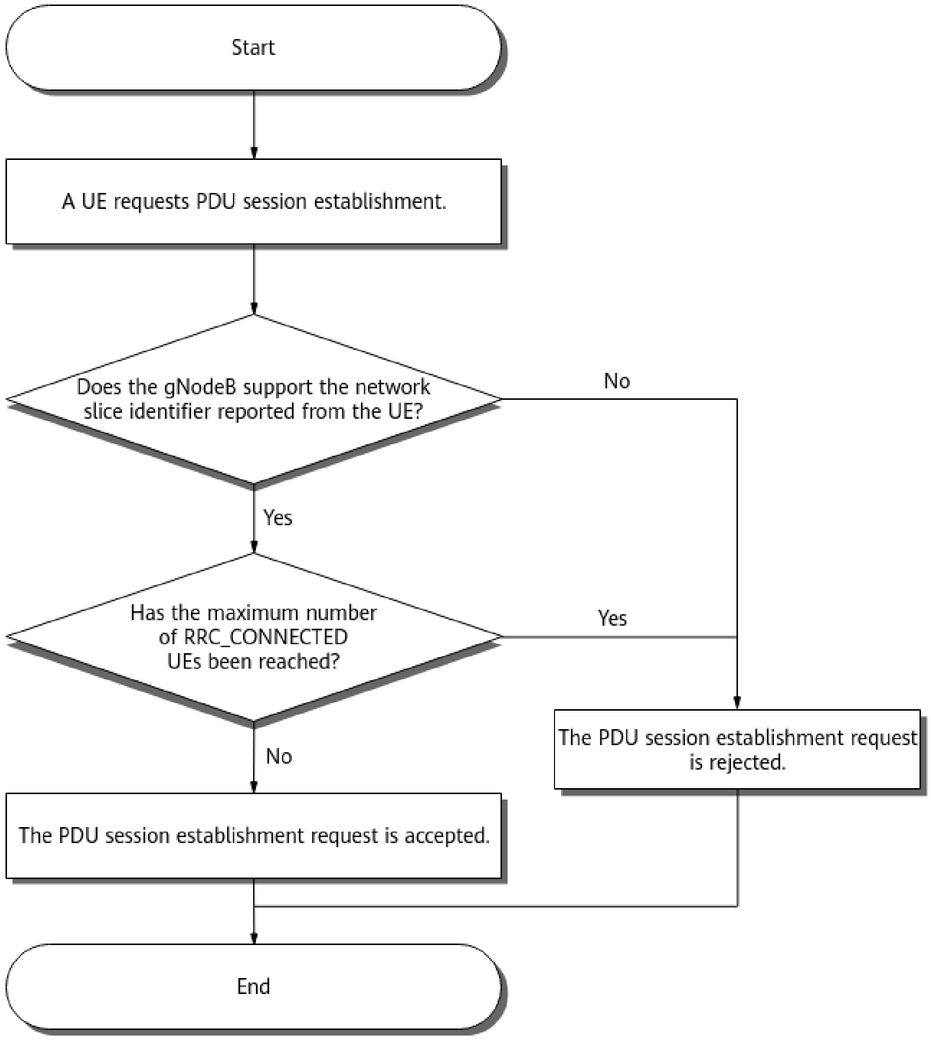
Figure 4-3 Network slice awareness in the radio bearer



4.1.2 Network Slice Admission Control

Following network slice deployment, the gNodeB performs admission control for slice UEs that access a cellular network, for the purpose of network slice isolation. Specifically, the gNodeB does so based on the network slices used for UEs and the number of RRC_CONNECTED UEs allowed by the corresponding network slice group. **Figure 4-4** shows the process.

Figure 4-4 Network slice admission control



Admission Control Based on Network Slice Identifiers

During a network access or handover procedure for a UE, the gNodeB compares the network slices used for the UE with those specific to a tracking area on the gNodeB (see **Configuring a Network Slice Specific to a Tracking Area on the gNodeB**). Based on the comparison result, the gNodeB decides whether to proceed with the network access or handover procedure. In this way, the gNodeB performs admission control based on network slice identifiers.

The **ADMISSION_BASED_ON_NS_ID_SW** option of the **NRCellAlgoSwitch.*NetworkSliceAlgoSwitch*** parameter specifies whether to enable this function.

NOTICE

Network slice deployment must be completed before you enable admission control based on network slice identifiers. Otherwise, admission control based on network slice identifiers cannot take effect.

Admission control based on network slice identifiers can be performed in the following scenarios:

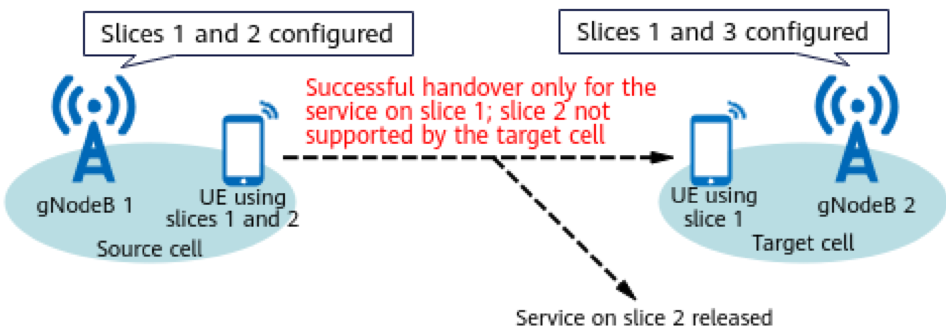
- PDU session establishment/modification and RRC connection resume/reestablishment:

  The gNodeB compares the S-NSSAI contained in a PDU session establishment request with the S-NSSAIs configured on the gNodeB. If the S-NSSAI in the request is the same as any of the S-NSSAIs, a PDU session is allowed to be established for the UE; otherwise, the request is rejected with the cause "Slice(s) not supported" or "Resources not available for the slice(s)", and the PDU session establishment fails.
- Intra-base-station and Xn-based handovers:

  The source cell obtains network slice configurations of a target cell. Then, the source cell checks whether a UE's network slices overlap with the network slices supported by the target cell. If they do, the UE initiates a handover request in the source cell and the gNodeB performs admission control based on network slice identifiers in the target cell. If they do not, the UE does not initiate a handover request in the source cell.
  - If the target cell supports all of the UE's network slices, the handover procedure is performed.
  - If the target cell supports some of the UE's network slices, the following applies: In a necessary handover, the handover procedure is performed and unsupported network slice services are released; in an unnecessary handover, the handover procedure is not performed.
- NG-based handovers:

  The source cell cannot obtain network slice configurations of a target cell. As such, a UE directly initiates a handover request in the source cell and the gNodeB performs admission control based on network slice identifiers in the target cell.
  - If the target cell supports all of the UE's network slices, the handover procedure is performed.
  - If the target cell supports some of the UE's network slices, the handover procedure is performed and unsupported network slice services are released, as shown in **Figure 4-5**.

    Figure 4-5 NG-based handovers when some of the UE's network slices are supported

  - If the target cell does not support any of the UE's network slices, the target cell notifies the source cell of a handover preparation failure, and the handover procedure stops.

If a network slice is supported by the source cell but not by a target cell, handovers to the target cell will be blocked for all UEs that use this network slice for a duration specified by the **gNBMobilityCommParam.*SliceNotSupportedTimer*** parameter.

NOTE

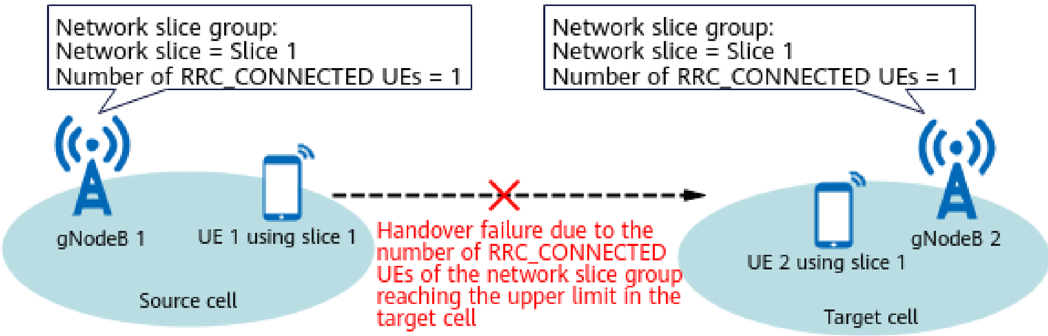
For details about the definitions of necessary handover and unnecessary handover, see *SA Mobility Management in Connected Mode*.

Admission Control Based on the Number of RRC_CONNECTED UEs

Network slicing supports admission control based on the number of RRC_CONNECTED UEs. For details about its principle and configuration, see **4.1.3 NR Cell Resource Management Between Network Slices**. After allowing a UE to establish a PDU session based on S-NSSAI judgment, the gNodeB further checks whether the maximum number of RRC_CONNECTED UEs has been reached in the network slice group to which the UE belongs. If the maximum number of RRC_CONNECTED UEs has not been reached, the gNodeB allows the UE to establish a PDU session; otherwise, the gNodeB rejects the PDU session establishment request.

With handovers, the gNodeB performs admission control based on the number of RRC_CONNECTED UEs in a target cell. **Figure 4-6** shows how this function works in an inter-base-station handover. In this example, if a UE using a network slice fails to be handed over from the source cell to a target cell because the maximum number of RRC_CONNECTED UEs configured in the target cell has been reached, handovers to the target cell will be blocked for all UEs that use this network slice for a duration specified by the **gNBMobilityCommParam.*ResNotAvailForSliceTimer*** parameter.

Figure 4-6 Admission control based on the number of RRC_CONNECTED UEs in an inter-base-station handover



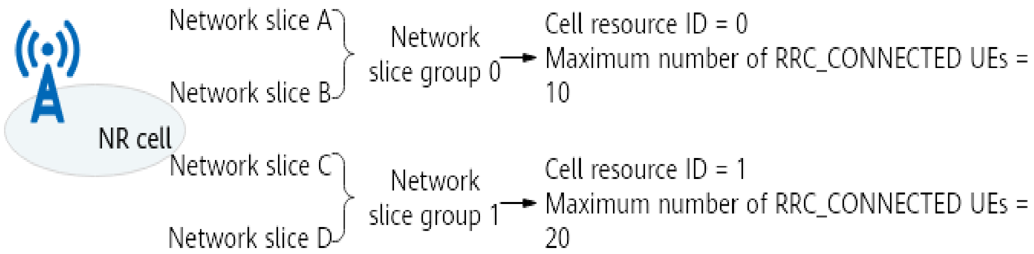
4.1.3 NR Cell Resource Management Between Network Slices

The NR cell resource management between network slices function is designed for resource isolation and UE admission control. NR cell resource management controls the number of RRC_CONNECTED UEs. The function associates S-NSSAIs to network slice groups and specifies the maximum number of RRC_CONNECTED UEs for these groups, thereby controlling UE access and handover.

The configuration items for NR cell resource management between network slices include: NR cell resources, network slice groups, and slices in network slice groups.

Figure 4-7 depicts an example. NR cell resource management between network slices is enabled by selecting the **ENHANCED_NETWORKSLICE_SW** option of the **NRCellAlgoSwitch.*NetworkSliceAlgoSwitch*** parameter.

Figure 4-7 NR cell resource management between network slices



The **NRCellRes** MO is used to configure NR cell resources, the **NRCellNsGrp** MO is used to configure a network slice group, and the **NRCellNsGrpNs** MO is used to configure a network slice in a network slice group. For details, see **4.4.1.1 Data Preparation**.

One cell supports a maximum of 32 network slice groups. The network slices configured in each group must belong to the same operator. In multi-operator scenarios where an NR cell is shared among operators, one or more network slice groups can be configured for each operator. Note that the total number of network slice groups of all operators must be less than or equal to 32.

It is good practice to set the maximum number of RRC_CONNECTED UEs (specified by **NRCellRes.*MaxRrcConnUserNum***) allowed in a network slice group of an NR cell according to the RBs occupied by a network slice group of an NR DU cell. For details on planning RBs occupied by a network slice group of an NR DU cell, see **4.1.4 NR DU Cell Resource Management Between Network Slices**.

In NSA and SA hybrid networking, resource sharing between network slice groups and RFSP groups is introduced to minimize configuration specifications and enable unified management of both. This function is supported only in low frequency bands. This function binds a network slice group and an RFSP group for the same operator to an NR cell resource group, allowing the maximum number of RRC_CONNECTED UEs to be shared between them. However, the gNodeB cannot provide differentiated guarantees for the network slice group and RFSP group that are bound to the same NR cell resource group. If differentiated guarantees are required, you are advised to bind the two groups to separate NR cell resource groups.

NOTE

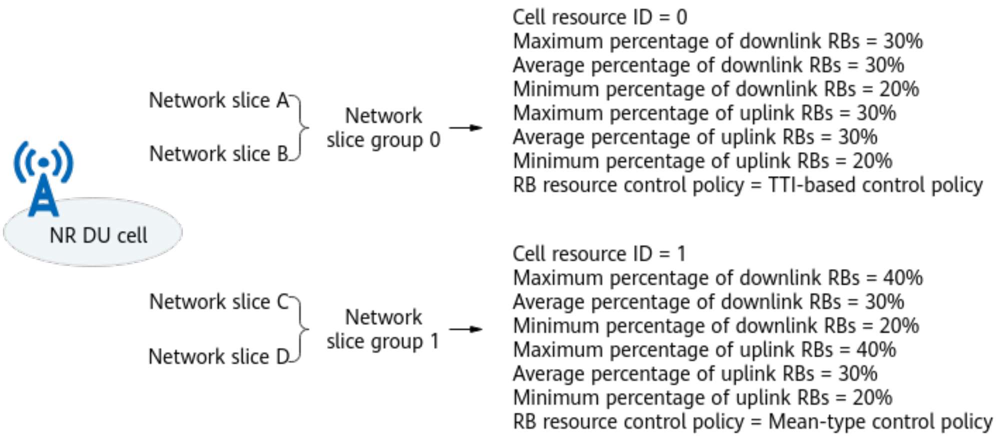
In NSA and SA hybrid networking, the **gNBOperator**.***NrNetworkingOption*** parameter must be set to **SA_NSA** and the **NRDUCellOp**.***NrNetworkingOption*** parameter for the cell must be set to **SA_NSA** or **INVALID**. For details, see *Cell Management*.

4.1.4 NR DU Cell Resource Management Between Network Slices

The NR DU cell resource management between network slices function is designed to satisfy SLA requirements of different services and ensure user experience. NR DU cell resource management controls the PUSCH RBs (excluding those for the PRACH and PUCCH) and PDSCH RBs (excluding those for the SSB, SIB, paging, RAR, and Msg4) available to a cell. This function uses network slice groups to dynamically control the resources occupied by network slices, thus ensuring SLAs of network slices.

NR DU cell resource management between network slices takes effect when both the **ENHANCED_NETWORKSLICE_SW** option of the **NRCellAlgoSwitch.*NetworkSliceAlgoSwitch*** parameter and the **RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch.*NetworkSliceAlgoSwitch*** parameter are selected. Resources are managed by configuring network slice groups, slices in the groups, and NR DU cell resources (percentage of RBs, start RB position, and RB resource control policy). **Figure 4-8** shows NR DU cell resource management between network slices.

Figure 4-8 NR DU cell resource management between network slices



NOTE

Refer to *Admission Control* to obtain the principles of service admission performed after NR DU cell resources are configured for network slices.

NR DU cell resource management between network slices includes basic resource management and enhanced resource management.

4.1.4.1 Basic Resource Management

The basic configuration items for NR DU cell resource management between network slices include: network slice groups, network slices in network slice groups, and NR DU cell resources. The RB resource control policy of an NR DU cell can be a TTI-based control policy or a mean-type control policy.

4.1.4.1.1 Configuring a Network Slice Group

The **NRDUCellNsGrp** MO is used to configure a network slice group. For details, see **4.4.1.1 Data Preparation**.

One cell supports a maximum of 32 network slice groups. The network slices configured in each group must belong to the same operator. In multi-operator scenarios where an NR cell is shared among operators, one or more network slice groups can be configured for each operator. Note that the total number of network slice groups of all operators must be less than or equal to 32.

In high-frequency scenarios, the network slice groups having the same value of **NRDUCellNsGrp.*NsGrpId*** must use the consistent network slice configurations across all cells within the same sector. Otherwise, these configurations specific to the aforesaid network slice groups cannot apply to any cell within the same sector. To facilitate the consistency check, the **DSP NRDUCELLNSGRP** command is used to display the consistency status of network slice group configurations across cells within the same sector.

4.1.4.1.2 Configuring Network Slices in a Network Slice Group

The **NRDUCellNsGrpNs** MO is used to configure network slices in a network slice group. For details, see **4.4.1.1 Data Preparation**.

4.1.4.1.3 Basic Resource Configuration Parameters for an NR DU Cell

The **NRDUCellRes** MO is used to configure the resources for an NR DU cell. The resource configuration items include the minimum percentage of RBs and its start frequency-domain position, average percentage of RBs, and maximum percentage of RBs. Note that the **NRDUCellRes**.***ResUsageObject*** parameter needs to be set to **NETWORK_SLICE_GROUP**.

NOTE

NR DU cell resource configuration takes effect on a per network slice group basis and is not affected by whether network slices are configured in the network slice group. If the minimum percentage of RBs is configured for a network slice group but there is no network slice in the group, the RBs specific to the configured minimum percentage cannot be used, resulting in resource waste.

Table 4-2 describes the basic configuration parameters for NR DU cell resources. In high-frequency scenarios, the network slice groups having the same value of **NRDUCellNsGrp.*NsGrpId*** must use the consistent settings of **Downlink RB Max Ratio**, **Uplink RB Max Ratio**, **Downlink RB Average Ratio**, **Uplink RB Average Ratio**, **Downlink RB Min Ratio**, and **Uplink RB Min Ratio** across all cells within the same sector. If any of these RB-ratio-related settings is inconsistent, these settings specific to the aforesaid network slice groups cannot apply to any cell within the same sector.

Table 4-2 Basic configuration parameters for NR DU cell resources

<table>
<tr><th>Parameter Name</th><th>Parameter ID</th><th>Description</th></tr>
<tr><td>Downlink RB Max Ratio</td><td>NRDUCellRes.DlRbMaxRatio</td><td rowspan="2">Indicates the maximum percentage of the cell's available PDSCH or PUSCH RBs that can be used by UEs in a network slice group. The RBs within the RB quota specified by the parameter are not preferentially used by these UEs.
Resources scheduled for DTX retransmissions are not restricted by the NRDUCellRes.DlRbMaxRatio or NRDUCellRes.UlRbMaxRatio parameter.
If there are multiple resource groups, the maximum percentage of RBs available for UEs in a resource group is not restricted by the configured maximum percentage of RBs. Instead, it is restricted by Min{Maximum percentage of RBs, 100% – Sum of the minimum percentages of RBs in other resource groups}.</td></tr>
<tr><td>Uplink RB Max Ratio</td><td>NRDUCellRes.UlRbMaxRatio</td></tr>
</table>

Parameter Name	Parameter ID	Description
Downlink RB Average Ratio	**NRDUCellRes.*DlRbAverageRatio***	Indicates the percentage of the cell's available PDSCH or PUSCH RBs that can be preferentially used by UEs in a network slice group. When not used by these UEs, the RBs within the RB quota specified by the parameter can be used by UEs outside the group. In the uplink, the **NRDUCellUlSch.*UlRbResSchPriority*** parameter specifies the scheduling priority policy of PUSCH RBs for UEs in the network slice group. Details are as follows: ● If this parameter is set to **NOT_CONFIG**, the priority of UEs in the network slice group is higher than that of UEs processing special services and lower than that of signaling messages. ● If this parameter is set to **HIGH_PRIORITY**, the priority of UEs in the network slice group is higher than that of signaling messages. ● If this parameter is set to **LOW_PRIORITY**, the priority of UEs in the network slice group is higher than that of UEs processing common data services and lower than that of UEs processing special services. Special services are non-common data services, such as the voice over NR (VoNR) service and URLLC service.
Uplink RB Average Ratio	**NRDUCellRes.*UlRbAverageRatio***	
Downlink RB Min Ratio	**NRDUCellRes.*DlRbMinRatio***	Indicates the minimum percentage of the cell's available PDSCH or PUSCH RBs that can be used by UEs in a network slice group. The RBs within the RB quota specified by the parameter are dedicated to these UEs.
Uplink RB Min Ratio	**NRDUCellRes.*UlRbMinRatio***	

Parameter Name	Parameter ID	Description
Downlink Min RB Start Position	**NRDUCellRes.*DlMinRbStartPos***	Indicates the start position of the minimum PDSCH or PUSCH RB quota specified by **NRDUCellRes.*DlRbMinRatio*** or **NRDUCellRes.*UlRbMinRatio*** (that is, the minimum percentage of the cell's available PDSCH or PUSCH RBs that can be used by UEs in a network slice group). ● For the downlink, the parameter can only be set to **0** (indicating the lower end of the cell bandwidth) and configured for only one network slice group of a cell. ● For the uplink, the parameter can only be set to **0** (indicating the lower end of the PUSCH frequency domain) and configured for only one network slice group of a cell.
Uplink Min RB Start Position	**NRDUCellRes.*UlMinRbStartPos***	
Resource Strategy Type	**NRDUCellRes.*ResStrategyType***	Indicates the RB resource control policy for a resource usage object in the NR DU cell. ● If this parameter is set to **TTI_ONLY**, RBs are scheduled in each TTI for a network slice group. For details, see **TTI-based Control Policy**. This policy is supported only in low frequency bands. ● If this parameter is set to **MEAN_ONLY**, RBs are scheduled based on the average PRB usage of UEs in a network slice group. For details, see **Mean-type Control Policy**.

4.1.4.1.4 Configuration Rules of Basic Parameters for NR DU Cell Resources

The resources that can be used by UEs in a network slice group are determined first by the minimum percentage of RBs, then by the average percentage of RBs, and finally by the maximum percentage of RBs. The configuration rules for uplink and downlink RB percentages are the same. This section describes the configuration rules for the downlink RB percentage as an example. For the meanings of parameters ***DlRbMaxRatio***, ***UlRbMaxRatio***, ***DlRbAverageRatio***, ***UlRbAverageRatio***, ***DlRbMinRatio***, and ***UlRbMinRatio***, see **4.1.4.1.3 Basic Resource Configuration Parameters for an NR DU Cell**.

Basic Configuration Rules of Percentage-related Parameters

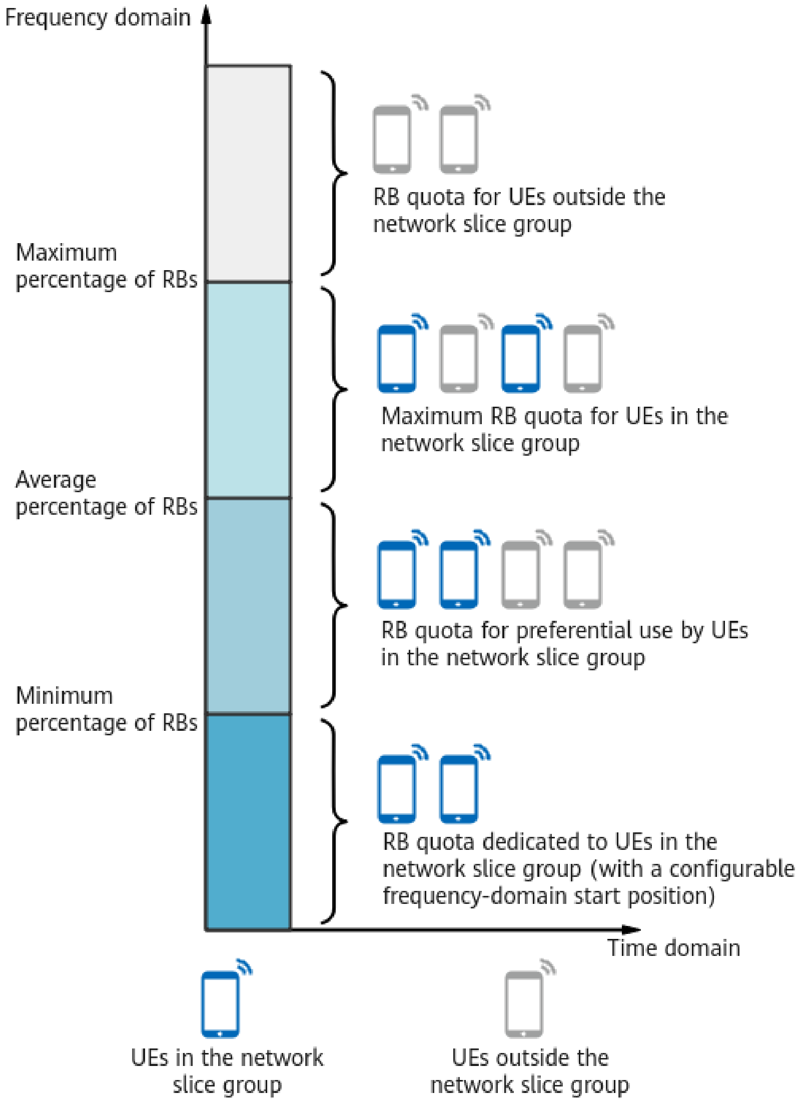
One or more of ***DlRbMaxRatio***, ***DlRbAverageRatio***, and ***DlRbMinRatio*** can be configured according to the network plan and the following requirements:

- If one or two parameters are configured, RB consumption is subject to the parameter descriptions in **Table 4-2**.

- If all of these parameters are configured, the value of ***DlRbMaxRatio*** must be greater than or equal to the value of ***DlRbAverageRatio***, which in turn must be greater than or equal to the value of ***DlRbMinRatio***. **Figure 4-9** shows the RB quotas specific to different percentages of RBs.
  - RBs within the quota specified by ***DlRbMinRatio*** are exclusively used by UEs in a network slice group.
  - RBs within the quota represented by "***DlRbAverageRatio*–*DlRbMinRatio***" are preferentially used by UEs in a network slice group and are available for UEs outside the group when not used.
  - RBs within the quota represented by "***DlRbMaxRatio*–*DlRbAverageRatio***" are not preferentially used by UEs in a network slice group but are shared with UEs outside the group.
- In an NR DU cell, the sum of the ***DlRbMinRatio*** values specified for all network slice groups must be less than or equal to 100%, and the same restriction applies to the ***DlRbAverageRatio*** values.

If ***DlRbMaxRatio***, ***UlRbMaxRatio***, ***DlRbAverageRatio***, ***UlRbAverageRatio***, ***DlRbMinRatio***, and ***UlRbMinRatio*** are all configured, these parameters cannot all be set to **255**.

Figure 4-9 RB configuration in an NR DU cell



Percentage-related Parameter Settings Specific to the Resource Allocation Granularity

The granularity for uplink and downlink resource allocation can be RBG or RB.

- When RBG-granularity scheduling is used, the maximum number of available RBs per network slice group and outside network slice groups must not be less than the number of RBs in one RBG. To fulfill this requirement, the following items must meet the specific configuration requirements listed in **Table 4-3**:
  - Sum of the minimum percentages of uplink or downlink RBs of all network slice groups in a cell
  - Sum of the average percentages of uplink or downlink RBs of all network slice groups in a cell
  - Maximum percentage of uplink or downlink RBs of a single network slice group in a cell

When the requirement is fulfilled, the actual number of RBs that can be scheduled will be less than the configured maximum number of RBs.

- There are no special requirements for using RB-granularity scheduling.

For details about uplink and downlink resource allocation modes, see *Uplink Scheduling* and *Downlink Scheduling*, respectively.

NOTE

An RBG may contain different numbers of RBs, depending on the applied bandwidth. For details, see section 5.3.2 "Transmission bandwidth configuration" in 3GPP TS 38.104 V15.5.0.

For details about uplink RBGs, see section 6.1.2.2.1 "Uplink resource allocation type 0" in 3GPP TS 38.214 V15.4.0.

For details about downlink RBGs, see section 5.1.2.2.1 "Downlink resource allocation type 0" in 3GPP TS 38.214 V15.4.0.

Table 4-3 Requirements for configuring percentages of RBs

RAT	Cell Bandwidth (MHz)	Number of RBs	Sum of the Minimum or Average Percentages of Uplink or Downlink RBs of All Network Slice Groups in a Cell[a]	Maximum Percentage of Uplink or Downlink RBs of a Single Network Slice Group
Low-frequency TDD	100	273	≤ 94	≥ 6
Low-frequency TDD	90	245	≤ 93	≥ 7
Low-frequency TDD	80	217	≤ 92	≥ 8
Low-frequency TDD	70	189	≤ 91	≥ 9
Low-frequency TDD	60	162	≤ 90	≥ 10
Low-frequency TDD	50	133	≤ 93	≥ 7
Low-frequency TDD	40	106	≤ 92	≥ 8

RAT	Cell Bandwidth (MHz)	Number of RBs	Sum of the Minimum or Average Percentages of Uplink or Downlink RBs of All Network Slice Groups in a Cell[a]	Maximum Percentage of Uplink or Downlink RBs of a Single Network Slice Group
Low-frequency TDD	30	78	≤ 89	≥ 11
Low-frequency TDD	20	51	≤ 92	≥ 8
High-frequency TDD	100	66	Downlink: ≤ 90 Uplink[b]: ≤ 65	≥ 10
High-frequency TDD	200	132	Downlink: ≤ 90 Uplink[b]: ≤ 65	≥ 10
a: If the average percentage of uplink or downlink RBs is configured for a network slice group, only the sum of the average percentages of uplink or downlink RBs of all network slice groups in a cell must meet the specific configuration requirements in this column. b: When PUSCH and PRACH FDM is enabled (by setting **NRDUCellPrach**.***PuschPrachFdmSwitch*** to **BASIC**), the uplink requirement is adjusted to "≤ 90". For details about PUSCH and PRACH FDM, see *Uplink Scheduling*.				

4.1.4.1.5 Specifying an RB Resource Control Policy

An RB resource control policy can be either a TTI-based control policy or a mean-type control policy, aiming to satisfy specific requirements of different service types on latency and resources. The TTI-based control policy is recommended only for secure isolation scenarios or for low-latency and high-reliability services, and the mean-type control policy is recommended only in scenarios other than low-latency and high-reliability scenarios. The RB resource control policy is specified by the **NRDUCellRes.*ResStrategyType*** parameter. The uplink and downlink RB resource control policies work in the same way. This section describes only the downlink RB resource control policy. For the meanings of parameters ***DlRbMaxRatio***, ***UlRbMaxRatio***, ***DlRbAverageRatio***, ***UlRbAverageRatio***, ***DlRbMinRatio***, ***UlRbMinRatio***, ***DlMinRbStartPos***, and ***UlMinRbStartPos***, see **4.1.4.1.3 Basic Resource Configuration Parameters for an NR DU Cell**.

TTI-based Control Policy

With this type of control policy, scheduling control is performed in each TTI for a network slice group. This policy takes effect when the

NRDUCellRes.*ResStrategyType* parameter is set to **TTI_ONLY**. A maximum of 6 resource groups with the TTI-based control policy can be configured for a cell. This policy is supported only in low frequency bands.

- If ***DlRbMinRatio*** is configured, the minimum number of downlink RBs available for UEs in the network slice group must be greater than or equal to the number corresponding to the configured value.
- If ***DlRbAverageRatio*** is configured, the number of downlink RBs that can be preferentially used by UEs in the network slice group must be greater than or equal to the number corresponding to the configured value. Note that, in this case, the following requirements must be met: ***DlRbMaxRatio*** is also configured, and ***DlRbAverageRatio*** is equal to ***DlRbMaxRatio***.
- If ***DlRbMaxRatio*** is configured, the maximum number of downlink RBs available for UEs in the network slice group cannot be greater than Min{***DlRbMaxRatio***, 100% – Sum of ***DlRbMinRatio*** for UEs in other resource groups}.

DlMinRbStartPos and ***UlMinRbStartPos*** can be configured only when the TTI-based control policy is used. These two parameters are used to mitigate interference from neighboring cells. The local cell and interfering neighboring cells must have the same values of ***DlMinRbStartPos*** and ***UlMinRbStartPos***.

The TTI-based control policy is recommended only for secure isolation scenarios or for low-latency and high-reliability services. If this policy is used, there must be a specific maximum number of RRC_CONNECTED UEs allowed by a network slice group (specified by **NRCellRes.*MaxRrcConnUserNum***) to meet the SLA requirements of UEs in the group. For details, contact Huawei technical support.

Mean-type Control Policy

When the mean-type control policy is used, the number of RBs available for UEs in a network slice group is not subject to the TTI-based control policy. Instead, RBs are scheduled based on the average PRB usage of these UEs (for example, **N.PRBUsed.UL.NsGrp1.Avg**/**N.PRB.UL.PUSCH.Avail.Avg** for uplink PRBs in network slice group 1). The mean-type control policy takes effect when **NRDUCellRes.*ResStrategyType*** is set to **MEAN_ONLY**.

- If the average PRB usage of UEs in the network slice group in the current TTI is less than or equal to ***DlRbAverageRatio***, RBs within the quota specified by ***DlRbAverageRatio*** are preferentially used by UEs in the network slice group in the next TTI.
- If the average PRB usage of UEs in the network slice group in the current TTI is greater than ***DlRbAverageRatio*** but less than or equal to Min{***DlRbMaxRatio***, 100% – Sum of ***DlRbMinRatio*** for UEs in other network slice groups}, RBs within the quota specified by ***DlRbAverageRatio*** are not preferentially used by UEs in the network slice group in the next TTI.
- If the average PRB usage of UEs in the network slice group in the current TTI is greater than Min{***DlRbMaxRatio***, 100% – Sum of ***DlRbMinRatio*** for UEs in other network slice groups}, UEs in the network slice group are not scheduled in the next TTI.

The mean-type control policy is recommended only in scenarios other than low-latency and high-reliability scenarios. ***DlRbAverageRatio*** can be configured to ensure the resources of network slice groups that are used in scenarios other than low-latency and high-reliability scenarios. As such, you are advised not to configure ***DlRbMinRatio*** when the mean-type control policy is used. If it is configured when the mean-type control policy is used, the RB usage of the cell is affected.

The difference between the percentage of RBs occupied by UEs in a network slice group and the configured RB percentage for the slice group may exceed 1% in low-frequency scenarios, when all of the following conditions are met:

- Sum of ***DlRbAverageRatio*** for multiple mean-type slice groups is equal to 100%.
- A sufficient number of UEs in each slice group perform full-buffer packet injection.
- The values of ***DlRbAverageRatio*** configured for these multiple slice groups differ greatly (by more than 70%).

Under these same conditions, the difference in high-frequency scenarios becomes even greater, as the number of available RBs decreases due to the beam limit.

4.1.4.2 Enhanced Resource Management

Beyond basic resource management, NR DU cell resource management between network slices provides enhanced functions in low-frequency scenarios, including downlink type 1 scheduling, uplink scheduling on a per RB basis, single-UE multi-slice RB management policy, resource sharing between network slice groups and RFSP groups, operator-specific resource sharing enhancement, interference randomization, and user-rate-based dynamic network slice resource management. These enhanced resource management functions accurately allocate resources in a more refined manner, minimizing resource waste and improving user experience. The parameters involved in these functions are ***DlRbMaxRatio***, ***UlRbMaxRatio***, ***DlRbAverageRatio***, ***UlRbAverageRatio***, ***DlRbMinRatio***, ***UlRbMinRatio***, ***DlMinRbStartPos***, and ***UlMinRbStartPos***. For their meanings, see **4.1.4.1.3 Basic Resource Configuration Parameters for an NR DU Cell**.

4.1.4.2.1 Downlink Type 1 Scheduling

By default, resources are allocated on a per RBG basis in downlink scheduling. If the RBs contained in an RBG outnumber the RB quota specified by ***DlRbMaxRatio*** for a network slice group with the TTI-based control policy, the RBs actually used will fall beyond the scope of configured RBs. As a result, RBs for UEs outside the group or in other groups are preempted, negatively impacting their user experience. Against this background, the downlink type 1 scheduling function has been introduced to accurately allocate downlink resources and ensure user experience. This function is controlled by the **DL_TYPE1_SCH_SW** option of the **NRDUCellRes.*SchedulingPolicySwitch*** parameter and can be enabled only when the **NRDUCellRes.*ResStrategyType*** parameter is set to **TTI_ONLY**. This function is supported only in low frequency bands.

If (1) downlink type 1 scheduling is enabled, and (2) the RBs contained in an RBG outnumber the RB quota specified by ***DlRbMaxRatio*** for a network slice group with the TTI-based control policy, then the gNodeB will allocate resources on a per four-RB basis. (If the RB quota specified by ***DlRbMaxRatio*** includes fewer than four RBs, four RBs are allocated.) In this case, the statistical average RBs used by a network slice group (for example, **N.PRBUsed.DL.NsGrp1.Avg** for network slice group 1) may still outnumber the RB quota specified by ***DlRbMaxRatio***. This means that the average downlink PRBs used by a network slice group may outnumber the average downlink PRBs available for the group.

4.1.4.2.2 Uplink Scheduling on a per RB Basis

By default, resources are allocated on a per four-RB basis in uplink scheduling. If the RB quota specified by ***UlRbMaxRatio*** includes fewer than four RBs for a network slice group with the TTI-based control policy, the RBs actually used may fall beyond the scope of configured RBs. As a result, RBs for UEs outside the group or UEs in other groups may be preempted. If this is the case, user experience of UEs outside the group or UEs in other groups deteriorates. Against this background, the uplink scheduling on a per RB basis function has been introduced to accurately allocate uplink resources and ensure user experience. This function is controlled by the **UL_1RB_SCH_SW** option of the **NRDUCellRes.*SchedulingPolicySwitch*** parameter and can be enabled only when the **NRDUCellRes.*ResStrategyType*** parameter is set to **TTI_ONLY**. This function is supported only in low frequency bands.

If the RB quota specified by **UlRbMaxRatio** includes fewer than four RBs for a network slice group with the TTI-based control policy, the gNodeB enabled with this function can allocate resources through resource scheduling on a per RB basis to ensure service continuity. In this case, the statistical average RBs used by a network slice group (for example, **N.PRBUsed.UL.NsGrp1.Avg** for network slice group 1) may still outnumber the RB quota specified by ***UlRbMaxRatio***. This means that the average uplink PRBs used by a network slice group may outnumber the average uplink PRBs available for the group. For example, when the **FAR_UE_MIN_RB_CONTROL_SW** or **UL_1RB_SCH_ENH_SW** option of the **NRDUCellPusch.*PuschPerformanceSwitch*** parameter is selected, the statistical average RBs used by a network slice group may outnumber the RB quota specified by the maximum percentage of RBs for the group.

4.1.4.2.3 Single-UE Multi-Slice RB Management Policy

When a single UE initiates multiple network slice services, RB management for the UE does not distinguish between network slices, whether in uplink or downlink scheduling. This leads to inaccurate resource allocation and negatively affects user experience. Against this background, the single-UE multi-slice RB management policy has been introduced to accurately allocate resources and ensure SLAs for different services. This policy is controlled by the **SINGLEUE_MULTISLICE_RB_CTRL_SW** option of the **NRDUCellAlgoSwitch.*NetworkSliceAlgoSwitch*** parameter. This policy is supported only in low frequency bands.

- When the single-UE multi-slice RB management policy is disabled, RB management for a single UE does not distinguish between network slices, whether in uplink or downlink scheduling. That is, all the RBs used by the UE are counted into the RBs used by the network slice group corresponding to the bearer that is configured with a slice resource group and has the highest logical channel priority. For details about the meaning and configuration of the logical channel priority, see *QoS Management*.
- When the single-UE multi-slice RB management policy is enabled, RB scheduling adheres to the rules described in **Table 4-4**.

Table 4-4 RB scheduling mechanism of the single-UE multi-slice RB management policy

Scheduling Direction	Service Type	RB Statistics Mechanism
Uplink	One network slice service initiated by a single UE[a]	The RBs used by the network slice service of the UE are counted into the RBs used by the network slice group to which the network slice belongs.
	Multiple network slice services concurrently initiated by a single UE	All the RBs used by the UE are counted into the RBs used by the network slice group corresponding to the bearer that is configured with a slice resource group and has the highest logical channel priority.
Downlink	One network slice service initiated by a single UE	The RBs used by the network slice service of the UE are counted into the RBs used by the network slice group to which the network slice belongs.
	Multiple network slice services concurrently initiated by a single UE	• RBs are scheduled on a per network slice basis.[b] • If one of the network slices belongs to a network slice group with the TTI-based control policy for which downlink type 1 scheduling takes effect, then downlink type 1 scheduling[c] is adopted for the UE.

a: When a single UE initiates only one network slice service in the uplink, network slice service bearers must belong to different logical channel groups. For the configuration related to logical channel groups, see the descriptions of QoS guarantee on the gNodeB side in the section on QoS guarantee for uplink scheduling in *QoS Management*.

b: The RBs used by a network slice service of the UE are counted into the RBs used by the network slice group to which the network slice belongs.

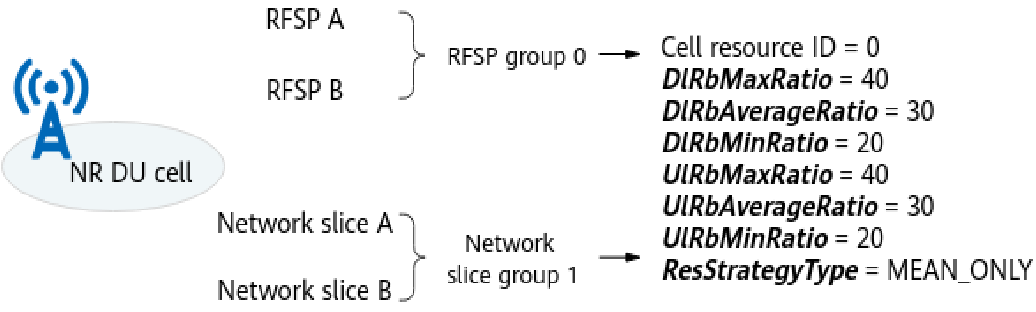
c: In downlink type 1 scheduling, a segment of consecutive RBs are allocated for the UE; that is, the RBs for multiple bearers of the UE are consecutive. For details about resource allocation in downlink type 1 scheduling, see section 5.1.2.2.2 "Downlink resource allocation type 1" in 3GPP TS 38.214 V15.4.0.

When the single-UE multi-slice RB management policy is enabled, a maximum of one resource group with the TTI-based control policy can be configured, and ***DlMinRbStartPos*** must be set to **65535**.

4.1.4.2.4 Resource Sharing Between Network Slice Groups and RFSP Groups

In NSA and SA hybrid networking, resource sharing between network slice groups and RFSP groups is introduced to minimize configuration specifications and enable unified management of both. This function is supported only in low frequency bands. When NR DU cell resource management between network slices and RFSP-group-based RB resource control are both enabled and the **NRDUCellRes.*ResUsageObject*** parameter for the NR DU cell resource group is set to **NS_RFSP_GROUP**, resource sharing between network slice groups and RFSP groups takes effect. As shown in **Figure 4-10**, this function binds a network slice group and an RFSP group for the same operator to an NR DU cell resource group, whose configurations are shared for scheduling UEs in both groups. In this case, the **NRDUCellRes.*ResStrategyType*** parameter (specifying the RB resource control policy) can only be set to **MEAN_ONLY**.

Figure 4-10 Configuration example for resource sharing between network slice groups and RFSP groups



NOTE

For details about RFSP-group-based RB resource control and RFSP groups, see section "RFSP-Group-based RB Resource Control" in *Resource Control in NSA Networking*.

"Resource sharing between network slice groups and RFSP groups" does not take effect when only "NR DU cell resource management between network slices" or "RFSP-group-based RB resource control" is enabled. In this case, only the enabled function takes effect.

The gNodeB cannot provide differentiated guarantees for the network slice group and RFSP group that are bound to the same NR DU cell resource group. To ensure differentiated service experiences for different UEs, you are advised to bind the two groups to separate NR DU cell resource groups.

4.1.4.2.5 Operator-specific Resource Sharing Enhancement

In RAN sharing with common carrier, operator-specific resource sharing enhancement is introduced to further refine resource allocation. This function, supported only in low frequency bands, ensures fair and flexible operator-specific resource sharing when NR DU cell resource management between network slices is enabled. The function involves intra-operator preferential sharing of idle resources and the inter-operator resource sharing policy. The inter-operator resource sharing policy takes effect only when intra-operator preferential sharing of idle resources is in effect. The function works the same way in both uplink and downlink cases, and the following uses the downlink case as an example. For meanings of ***DlRbAverageRatio*** and ***DlRbMinRatio***, see **4.1.4.1.3 Basic Resource Configuration Parameters for an NR DU Cell**. For details about RAN sharing with common carrier, see *Multi-Operator Sharing*.

This section specifically describes operator-specific resource sharing enhancement in SA networking. For information on operator-specific resource sharing enhancement in NSA networking, see *Resource Control in NSA Networking*. In

NSA and SA hybrid networking, "network slice group" mentioned in this section is equivalent to "network slice group or RFSP group".

As for some below mentioned concepts:

- Idle resources are resources that are not used by UEs in a network slice group within the range of ***DlRbAverageRatio*** minus ***DlRbMinRatio*** specified for the network slice group. If ***DlRbAverageRatio*** is set to **255**, there are no idle resources.
- Common resources are all remaining resources beyond the quota represented by the sum of the larger of either ***DlRbAverageRatio*** or ***DlRbMinRatio*** specified for all network slice groups (or all network slice groups and RFSP groups in NSA and SA hybrid networking) in a cell. If ***DlRbAverageRatio*** or ***DlRbMinRatio*** is set to **255**, ***DlRbAverageRatio*** or ***DlRbMinRatio*** is considered 0 in calculation of common resources.
- Operator-specific shares are the sum of the larger of either ***DlRbAverageRatio*** or ***DlRbMinRatio*** specified for all network slice groups (or all network slice groups and RFSP groups in NSA and SA hybrid networking) of an operator. If ***DlRbAverageRatio*** or ***DlRbMinRatio*** is set to **255**, ***DlRbAverageRatio*** or ***DlRbMinRatio*** is considered 0 in calculation of operator-specific shares.

Intra-Operator Preferential Sharing of Idle Resources

This function allocates idle resources preferentially to UEs served by the same operator and is controlled by the **NRDUCellNsMgmtConfig**.***NetSliceRfspGrpSharePol*** parameter.

- If this parameter is set to **SAME_OPERATOR_PREFERRED**, idle resources are preferentially shared among UEs served by the same operator. If there are remaining idle resources, they can be shared with UEs served by other operators according to the inter-operator resource sharing policy described in **Inter-Operator Resource Sharing Policy**.
- If this parameter is set to **FAIR_SHARING**, intra-operator preferential sharing of idle resources does not take effect, and idle resources are allocated based on UE scheduling priorities.

Inter-Operator Resource Sharing Policy

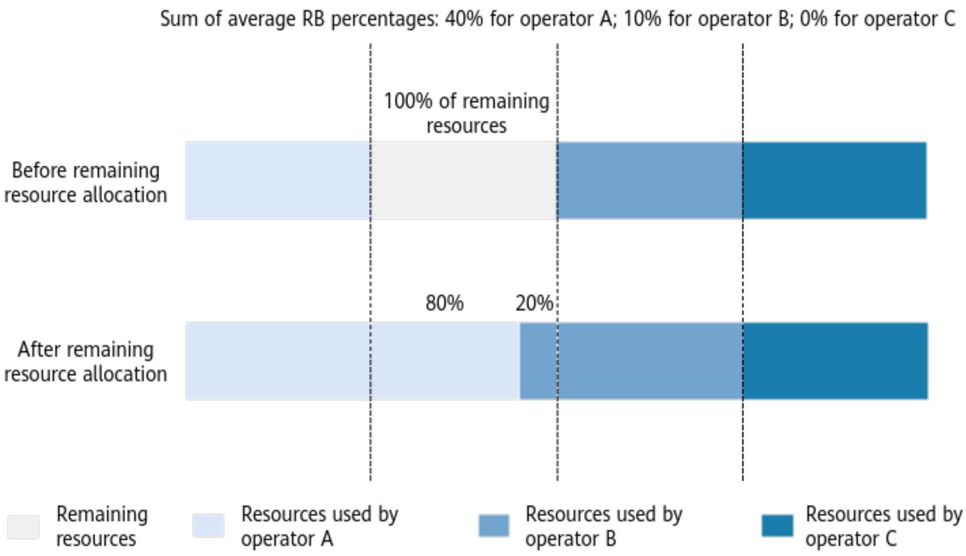
When the **NRDUCellNsMgmtConfig**.***NetSliceRfspGrpSharePol*** parameter is set to **SAME_OPERATOR_PREFERRED**, idle resources are preferentially shared among UEs served by the same operator. The remaining idle resources (if any) and common resources, altogether referred to as the remaining resources, can be shared with UEs served by other operators according to the specific inter-operator resource sharing policy. An inter-operator resource sharing policy can be based on either operator-specific shares or the number of active UEs, specified by the **NRDUCellNsMgmtConfig.*NetSliceRfspGrpPreemptPol*** parameter.

- If this parameter is set to **BASED_ON_OP_SHARE**, resource sharing based on operator-specific shares takes effect. The gNodeB allocates the remaining resources in proportion to each operator's share.

  As shown in **Figure 4-11**, if the sum of average RB percentages configured for all network slice groups of operators A, B, and C in a cell are 40%, 10%, and

0%, respectively, then: 80% (4/5) of the remaining resources can be preferentially used by operator A; 20% (1/5) can be preferentially used by operator B; there are no remaining resources that can be preferentially used by operator C.

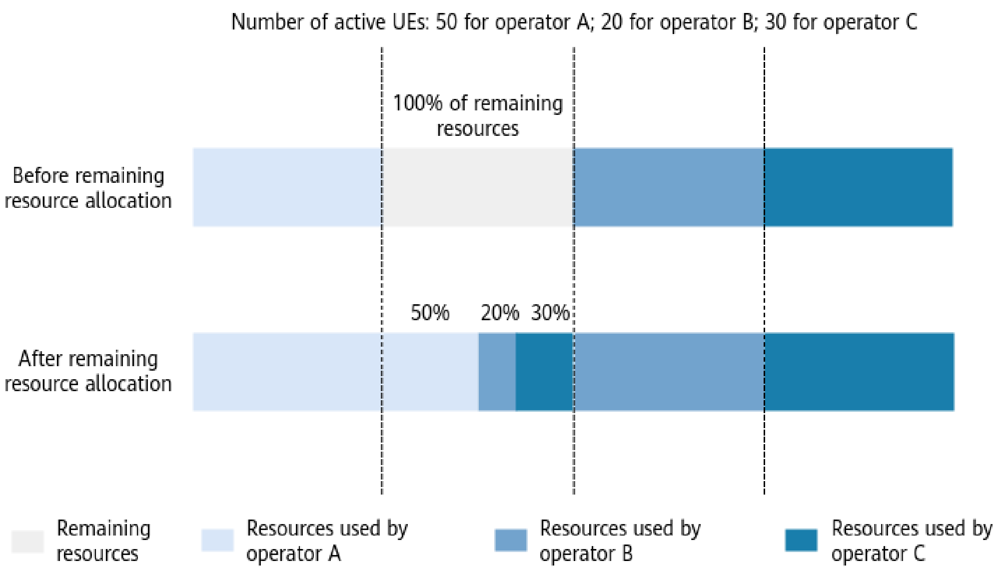
Figure 4-11 Example of resource sharing based on operator-specific shares



- If this parameter is set to **BASED_ON_OP_ACTIVE_USER**, resource sharing based on the number of active UEs takes effect. The gNodeB allocates the remaining resources in proportion to each operator's active UE number.

  As shown in **Figure 4-12**, if the numbers of active UEs served by operators A, B, and C in a cell are 50, 20, and 30, respectively, then: 50% (1/2) of the remaining resources can be preferentially used by operator A; 20% (1/5) can be preferentially used by operator B; 30% (3/10) can be preferentially used by operator C.

Figure 4-12 Example of resource sharing based on the number of active UEs



NOTE

An active UE is a UE in RRC_CONNECTED mode for which the buffer status report (BSR) is not empty.

When an inter-operator resource sharing policy takes effect and the **NRDUCellRes**.***ResUsageObject*** parameter is set to **NETWORK_SLICE_GROUP**, **NS_RFSP_GROUP**, or **RFSP_GROUP**, the sum of the following results must not exceed 100%. This applies to both uplink and downlink cases, and the following uses the downlink case as an example.

- Sum of ***DlRbAverageRatio*** in all records with ***DlRbAverageRatio*** set to a value other than **255**
- Sum of ***DlRbMinRatio*** in all records with ***DlRbAverageRatio*** set to **255** and ***DlRbMinRatio*** set to a value other than **255**

4.1.4.2.6 Interference Randomization

A network slice group configured with the TTI-based control policy supports interference randomization in scenarios requiring low latency and high reliability. That is, interference randomization is available within a fixed reserved resource range for the network slice group, reducing mutual interference between the victim cells running low-latency and high-reliability services within the fixed reserved resource range. This function is supported only in low frequency bands. For details about interference randomization in low-latency and high-reliability scenarios, see *URLLC*.

Interference randomization is enabled when the **FIX_RES_UL_INTRF_RANDOM_SCH_SW** and **FIX_RES_DL_INTRF_RANDOM_SCH_SW** options (controlling uplink and downlink interference randomization, respectively) of the **NRDUCellRes.*SchedulingPolicySwitch*** parameter are selected. You can select either or both of them. This function requires the following additional configurations:

- Uplink interference randomization requires the following additional configurations:
  - ***UlMinRbStartPos*** is set to **0**, indicating that the fixed reserved resources are allocated starting from the RB following the last RB occupied by the PUCCH.
  - ***UlRbMaxRatio*** and ***UlRbMinRatio*** have the same value.
- Downlink interference randomization requires the following additional configurations:
  - ***DlMinRbStartPos*** is set to **0**, indicating that the fixed reserved resources are allocated starting from RB 0.
  - ***DlRbMaxRatio*** and ***DlRbMinRatio*** have the same value.

4.1.4.2.7 User-Rate-based Dynamic Network Slice Resource Management

UEs running different services require different data rates. If the target data rates of UEs cannot be reached due to inappropriate settings of **Uplink RB Average Ratio** or **Downlink RB Average Ratio**, the service experience of these UEs cannot be guaranteed. To address this, the "user-rate-based dynamic network slice resource management" function is introduced, which is controlled by the

DYNAMIC_NETWORKSLICE_SW option of the **NRDUCellCommonSw.*ServiceDiffAlgoExtSwitch*** parameter. This function dynamically adjusts the average percentages of uplink and downlink RBs in NR DU cell resources. This function requires the **NRDUCellRes.*ResStrategyType*** parameter to be set to **MEAN_ONLY**. This function is supported only in low frequency bands.

When the aforesaid option is selected and **Uplink RB Average Ratio** or **Downlink RB Average Ratio** is configured for the **NRDUCellRes** MO, the gNodeB dynamically adjusts RB allocation based on the **NRDUCellQciBearer.*DynResUlTargetRate*** parameter (indicating the uplink target rate specific to dynamic resources) or the **NRDUCellQciBearer.*DynResDlTargetRate*** parameter (indicating the downlink target rate specific to dynamic resources) specified for UEs in a network slice group. In this situation, RBs are not preferentially allocated based on the configured **Uplink RB Average Ratio** or **Downlink RB Average Ratio**. Specifically, if the average percentage of uplink or downlink RBs estimated by the gNodeB for all network slice groups covered by an NR DU cell transmission reception point (TRP) is less than the configured **Uplink RB Average Ratio** or **Downlink RB Average Ratio**, then RBs are allocated to those network slice groups based on the configured **Uplink RB Average Ratio** or **Downlink RB Average Ratio**. For details about the NR DU cell TRP, see *Cell Management*.

The gNodeB reserves some RBs to ensure the basic service experience for UEs outside network slice groups. UEs in network slice groups do not have preferential access to these reserved RBs. For details, see **Table 4-5**.

Table 4-5 Number of RBs reserved for UEs outside network slice groups in cells of different bandwidths and RATs

RAT	Cell Bandwidth (MHz)	Number of RBs	Number of RBs Reserved for UEs Outside Network Slice Groups
Low-frequency TDD	100	273	16
Low-frequency TDD	90	245	16
Low-frequency TDD	80	217	16
Low-frequency TDD	70	189	16
Low-frequency TDD	60	162	16

RAT	Cell Bandwidth (MHz)	Number of RBs	Number of RBs Reserved for UEs Outside Network Slice Groups
Low-frequency TDD	50	133	8
Low-frequency TDD	40	106	8
Low-frequency TDD	30	78	8
Low-frequency TDD	20	51	4

4.2 Network Analysis

4.2.1 Benefits

The network slicing function enables multiple logical networks to run on an operator's physical network. This function provides 5G network services tailored to different requirements, reducing costs for operators and improves competitiveness for vertical industries.

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

- NR DU cell resource management between network slices:

  It is recommended that SR-based smart scheduling be enabled (by selecting the **SR_SMART_SCH_SW** option of the **NRDUCellPusch.*UlPuschAlgoSwitch*** parameter) when you enable NR DU cell resource management between network slices. If the former function is not enabled together with the latter one, the scheduling priorities of UEs outside slice groups may be higher than those of UEs inside slice groups, meaning that SLAs for UEs inside slice groups cannot be ensured.

  In high-frequency scenarios, the number of available beams in a slot is limited. In this context, if the minimum percentage of RBs (specified by the **NRDUCellRes.*DlRbMinRatio*** or **NRDUCellRes.*UlRbMinRatio*** parameter) is set to an excessively high value for a network slice group, the function has the following network impacts:

  – The resources available to UEs outside the network slice group are critically insufficient, causing repeated access attempts and frequent service drops for these UEs.

- If the carriers supported by a specific UE in the network slice group are fewer than the actual carriers in the sector, the RBs configured by the minimum RB percentage may fail to be reserved for this UE.
- The beam resources occupied by UEs fluctuate, resulting in an increased difference between the percentage of RBs used by UEs in each network slice group and the configured RB percentage.

- Uplink scheduling on a per RB basis:

  After uplink scheduling on a per RB basis is enabled, UEs in the slice groups are scheduled on a per RB basis in the uplink. In such a case, the average number of uplink PRBs used in a lightly-loaded slice group decreases (for example, the value of the **N.PRBUsed.UL.NsGrp0.Avg** counter decreasing for slice group 0), whereas the uplink throughput of a heavily-loaded slice group increases.

- Downlink type 1 scheduling:

  After downlink type 1 scheduling is enabled, the average number of downlink PRBs used in a slice group decreases (for example, the value of the **N.PRBUsed.DL.NsGrp0.Avg** counter decreasing for slice group 0).

  When multiple slice groups with the TTI-based control policy are configured and two or more of them are enabled with downlink type 1 scheduling, the number of downlink scheduling times and the number of requested CCEs increase.

- Operator-specific resource sharing enhancement:
  - Enabling intra-operator preferential sharing of idle resources has the following impacts: The downlink and uplink cell throughputs (**Cell Downlink Average Throughput (DU)** and **Cell Uplink Average Throughput (DU)**) fluctuate. For an operator with more RBs allocated after the function is enabled, the downlink and uplink UE throughputs (**User Downlink Average Throughput (DU)** and **User Uplink Average Throughput (DU)**) increase, improving user experience. For an operator with fewer RBs allocated after the function is enabled, the downlink and uplink UE throughputs decrease, negatively affecting user experience.
  - When the inter-operator resource sharing policy takes effect, the impacts are as follows: The downlink and uplink cell throughputs fluctuate. For an operator with a large proportion of resources allocated, the downlink and uplink UE throughputs increase, improving user experience. For an operator with a small proportion of resources allocated, the downlink and uplink UE throughputs decrease, negatively affecting user experience.

- User-rate-based dynamic network slice resource management:

  If the RBs required by the uplink/downlink target rates of UEs in a network slice group, based on dynamic resources, exceed the RBs indicated by the configured average percentages of uplink/downlink RBs, enabling user-rate-based dynamic network slice resource management will have the following network impacts:
  - If the target rate satisfaction level of the network slice group is low, user experience in the group improves.
  - If the cell load is medium or heavy (PRB usage exceeding 30%) and there are a large number of cell edge UEs in the group, the uplink and downlink cell throughputs (**Cell Downlink Average Throughput (DU)** and **Cell Uplink Average Throughput (DU)**) decrease.

The rate satisfaction level is calculated using the following formulas:

- The formula below assumes that the counters specific to an uplink minimum GBR of 3 Mbit/s are used:

  Uplink rate satisfaction level = (**N.Thp.UL.Flex.Index3** + ... + **N.Thp.UL.Flex.Index14**)/(**N.Thp.UL.Flex.Index0** + ... + **N.Thp.UL.Flex.Index14**)

- The formula below assumes that the counters specific to a downlink minimum GBR of 5 Mbit/s are used:

  Downlink rate satisfaction level = (**N.Thp.DL.Flex.Index2** + ... + **N.Thp.DL.Flex.Index14**)/(**N.Thp.DL.Flex.Index0** + ... + **N.Thp.DL.Flex.Index14**)

NOTE

Flexible performance counters (such as **N.Thp.UL.Flex.Index0**) can be configured using the **gNBFlexCounterObjCfg** MO. For details, see "Custom Object Type" in the "Object Type" section in *Performance Management*.

4.3 Requirements

4.3.1 Licenses

When network slice resource management needs to be enabled (by selecting the **ENHANCED_NETWORKSLICE_SW** option of the **NRCellAlgoSwitch.*NetworkSliceAlgoSwitch*** parameter), apply for the following license.

Feature ID	Feature Name	Model	Sales Unit
FOFD-050201	Network Slicing Introduction	NR0S00BNSL00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-operator preferential sharing of idle resources	**NRDUCellNsMgmtConfig**.***NetSliceRfspGrpSharePol*** set to **SAME_OPERATOR_PREFERRED**	*Network Slicing*	The inter-operator resource sharing policy (specified by the **NRDUCellNsMgmtConfig**.***NetSliceRfspGrpPreemptPol*** parameter) depends on intra-operator preferential sharing of idle resources.
Low-frequency TDD	NR DU cell resource management between network slices	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter	*Network Slicing*	Intra-operator preferential sharing of idle resources (with the **NRDUCellNsMgmtConfig**.***NetSliceRfspGrpSharePol*** parameter set to **SAME_OPERATOR_PREFERRED**) depends on "NR DU cell resource management between network slices".
High-frequency TDD	Network slicing enhancement	**ENHANCED_NETWORKSLICE_SW** option of the **NRCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter	*Network Slicing*	The dynamic RB control switch (that is, the **RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter) can be turned on only after the network slicing enhancement switch is turned on.
Low-frequency TDD	NR DU cell resource management between network slices	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter	*Network Slicing*	User-rate-based dynamic network slice resource management (controlled by the **DYNAMIC_NETWORKSLICE_SW** option of the **NRDUCellCommonSw**.***ServiceDiffAlgoExtSwitch*** parameter) can be enabled only after "NR DU cell resource management between network slices" is enabled.

4.3.2.2 Mutually Exclusive Functions

- Functions that are mutually exclusive with NR DU cell resource management between network slices (controlled by the **RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch.*NetworkSliceAlgoSwitch*** parameter)

  For LTE TDD and NR spectrum sharing, this mutually exclusive relationship applies when all of the following conditions are satisfied: (1) the cell is configured as a LampSite cell (with **NRDUCell**.***LampSiteCellFlag*** set to **YES**); (2) cell combination networking mode (with **NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**) is adopted; (3) DPS transmission mode (with **NRDUCellMultiTrp**.***DataTransMode*** set to **DPS_MODE**) is adopted.

RAT	Function Name	Function Switch	Reference
Low-frequency TDD High-frequency TDD	Operator-specific RB management	**RB_DYNAMIC_SHARING_SW** option of the **NRDUCellAlgoSwitch.*RanSharingAlgoSwitch*** parameter	*Multi-Operator Sharing*
Low-frequency TDD	Operator-specific limitation on the maximum number of available RBs	**MAX_RB_LIMIT_SW** option of the **NRDUCellAlgoSwitch.*RanSharingAlgoSwitch*** parameter	*Multi-Operator Sharing*
Low-frequency TDD	Operator-specific guarantee on the minimum number of available RBs	**MIN_RB_GUARANTEE_SW** option of the **NRDUCellAlgoSwitch.*RanSharingAlgoSwitch*** parameter	*Multi-Operator Sharing*
Low-frequency TDD	Cell Combination	**HYPER_CELL_COMBINE_MODE** option of the **NRDUCell.*NrDuCellNetworkingMode*** parameter selected with the **NRDUCellMultiTrp.*DataTransMode*** parameter set to **BASIC_MODE**	*Cell Combination*

RAT	Function Name	Function Switch	Reference
Low-frequency TDD High-frequency TDD	User PRB control for FWA	**PRB_CTRL_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*
Low-frequency TDD High-frequency TDD	Dynamic RB control for UEs requiring low latency and high reliability	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	*URLLC*
Low-frequency TDD	LTE TDD and NR spectrum sharing	**LTE_NR_TDD_SPCT_SHR_SW** option of the **NRDUCellAlgoSwitch**.***SpectrumCloudSwitch*** parameter	None
Low-frequency TDD	FSD transmission mode	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**, and **NRDUCellMultiTrp**.***DataTransMode*** set to **FSD_MODE**	*Hyper Cell*
High-frequency TDD	UL Grant-free	**GRANT_FREE_SW** option of the **NRDUCellAlgoSwitch**.***LowLatencySwitch*** parameter	None
High-frequency TDD	Low Latency High Reliability Level 2	**LOW_LAT_HIGH_RELIAB_LEVEL2_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	None

RAT	Function Name	Function Switch	Reference
High-frequency TDD	mmWave level-1 low jitter and high reliability	**L_JIT_H_RELIAB_MMW_LVL1_SW** option of the **NRDUCellAlgoSwitch**.***HighReliabilitySwitch*** parameter	None
High-frequency TDD	Low Latency High Reliab mmWave Adv	**LOW_LAT_HIGH_RELIAB_MMW_ADV_SW** option of the **NRDUCellFeatureSw**.***HighReliabilityExtSwitch*** parameter	None
High-frequency TDD	Submeter-level precise positioning	**AOA_UTDOA_HYBRID_LOCATION_SW** option of the **NRCellAlgoSwitch**.***LcsSwitch*** parameter	None
High-frequency TDD	mmWave Cell Combination	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL_COMBINE_MODE**	None
High-frequency TDD	Hyper Cell	**NRDUCell**.***NrDuCellNetworkingMode*** set to **HYPER_CELL**	None
High-frequency TDD	Cell coordination	**INTRA_GNB_DL_JT_SW** option of the **NRDUCELLALGOSWITCH**.***CompSwitch*** parameter	None
Low-frequency TDD	User-rate-based dynamic QCI resource management	**DYNAMIC_QCI_GROUP_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*QCI-based Resource Control*

- Functions that are mutually exclusive with the minimum downlink RB ratio (specified by **NRDUCellRes.*DlRbMinRatio***) or minimum uplink RB ratio (specified by **NRDUCellRes.*UlRbMinRatio***)

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	RedCap Introduction	**REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter	*RedCap*
Note that the following mutually exclusive relationships apply if RedCap Introduction is enabled in an NR TDD cell: ● If **NRDUCellRes.*DlMinRbStartPos*** is set to a value other than **65535**, **NRDUCellRes.*DlRbMinRatio*** cannot be set to a value greater than **10**. ● If **NRDUCellRes.*UlMinRbStartPos*** is set to a value other than **65535**, **NRDUCellRes.*UlRbMinRatio*** cannot be set to a value greater than **10**.			

- Functions that are mutually exclusive with the downlink minimum RB start position (specified by **NRDUCellRes.*DlMinRbStartPos***) or the uplink minimum RB start position (specified by **NRDUCellRes.*UlMinRbStartPos***)

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	Downlink RB restriction for the low-speed cell[a]	**NRDUCellHighSpeedMob**.***DlIntrfLimitRbSchRatio***	*High Speed Mobility*
Low-frequency TDD	Uplink RB restriction for the low-speed cell[b]	**NRDUCellHighSpeedMob**.***UlIntrfLimitRbSchRatio***	*High Speed Mobility*
a: If the downlink minimum RB start position is set to **0**, the downlink RB restriction for the low-speed cell must be set to **0**. b: If the uplink minimum RB start position is set to **0**, the uplink RB restriction for the low-speed cell must be set to **0**.			

- Functions that are mutually exclusive with intra-operator preferential sharing of idle resources (enabled by setting the **NRDUCellNsMgmtConfig**.***NetSliceRfspGrpSharePol*** parameter to **SAME_OPERATOR_PREFERRED**)

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	QCI group	**NRDUCellRes**.***ResUsageObject*** set to **QCI_GROUP**	*QCI-based Resource Control*

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	QCI-based resource control	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*QCI-based Resource Control*
Low-frequency TDD	User-rate-based dynamic network slice resource management	**DYNAMIC_NETWORKSLICE_SW** option of the **NRDUCellCommonSw**.***ServiceDiffAlgoExtSwitch*** parameter	*Network Slicing*

- Functions that are mutually exclusive with user-rate-based dynamic network slice resource management (controlled by the **DYNAMIC_NETWORKSLICE_SW** option of the **NRDUCellCommonSw**.***ServiceDiffAlgoExtSwitch*** parameter)

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	QCI-based resource control	**RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*QCI-based Resource Control*
Low-frequency TDD	Bi-Carrier	**BI_CARRIER_SW** option of the **NRDUCellFeatureSw**.***MultiCarrierSwitch*** parameter	None
Low-frequency TDD	Slice-based QoS scheduling	**SLICE_BASED_QOS_SCH_SW** option of the **NRDUCellAlgoSwitch**.***NetworkSliceAlgoSwitch*** parameter	None

RAT	Function Name	Function Switch	Reference
Low-frequency TDD	Service admission	**QOS_FLOW_ADMISSION_SW** option of the **NRDUCellAlgoSwitch**.***RacAlgoSwitch*** parameter	*Admission Control*
Low-frequency TDD	Operator-specific resource sharing enhancement	**NRDUCellNsMgmtConfig**.***NetSliceRfspGrpSharePol*** set to **SAME_OPERATOR_PREFERRED**	*Network Slicing*

4.3.2.3 Function Impacts

Network slice deployment:

- Functions related to MIMO:

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Co-MM	**INTRA_GNB_SU_COMM_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	*Co-MM (Low-Frequency TDD)*	The network slices configured for a cooperating cell must include UEs' network slices.

- Functions related to coordination:

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station UL CoMP	**INTRA_GNB_UL_COMP_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	The network slices configured for a cooperating cell must include UEs' network slices.
Low-frequency TDD	Intra-base-station DL CoMP	**INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch**.***CompSwitch*** parameter	*CoMP*	The network slices configured for a cooperating cell must include UEs' network slices.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-base-station downlink joint transmission	**INTER_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch.*CompSwitch*** parameter	None	The network slices configured for a cooperating cell must include UEs' network slices.

- Functions related to carrier management:

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch.*CaAlgoSwitch*** parameter	*Carrier Aggregation*	The network slices configured for an SCell must include UEs' network slices.
Low-frequency TDD	Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch.*CaAlgoSwitch*** parameter	*Carrier Aggregation*	The network slices configured for an SCell must include UEs' network slices.
High-frequency TDD	Downlink intra-band CA	**INTRA_BAND_CA_SW** and **INTRA_BAND_UL_CA_SW** options of the **NRDUCellAlgoSwitch.*CaAlgoSwitch*** parameter	*Carrier Aggregation*	The network slices configured for an SCell must include UEs' network slices.
Low-frequency TDD High-frequency TDD	Inter-FR CA	**INTER_FR_FR1TOFR2_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	The network slices configured for an SCell must include UEs' network slices.

- Basic functions:

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Frequency-based 5GC selection	**gNBCUNgTrackingArea.*gNBCuNgId*** and **gNBCUNgTrackingArea.*TrackingAreaId***	*Multi-Operator Sharing*	When both Network Slicing Introduction and frequency-based 5GC selection are enabled, there must be an NG interface for which **gNBCUNg.*CuNgType*** is set to **DEFAULT_CP** or **DEFAULT_CP_UP** among an operator's NG interfaces corresponding to the same **gNBCUNgTrackingArea.*TrackingAreaId*** value. Otherwise, the gNodeB cannot select a default AMF for a network slice UE if the UE does not report either the S-NSSAIs supported by the gNodeB or the Temp ID, resulting in access failures. In live network planning, if a **gNBCUNg** MO of the default type (**gNBCUNg.*CuNgType*** set to **DEFAULT_CP** or **DEFAULT_CP_UP**) is configured, there must be an available NG link under the **gNBCUNg** MO of the default type. Otherwise, services will be affected.

Network slice admission control:

- Functions related to mobility management:

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UE-number-based connected mode MLB	**INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch.*MlbAlgoSwitch*** parameter selected, and **NRCellMlb.*MlbTriggerMode*** set to **CELL_USER_NUM**	*Mobility Load Balancing in Connected Mode*	Assume that both network slice admission control and UE-number-based connected mode MLB are enabled and UEs in a network slice are selected for MLB. If the maximum number of UEs in a target cell has been reached, the network slice UEs cannot be handed over from the source cell to the target cell. To reduce possibility of a decrease in the handover success rate due to invalid handovers, the **gNBMobilityCommParam.*ResNotAvailForSliceTimer*** parameter must be set to a value greater than 20s. In this case, all UEs in the network slice will not be handed over to the target cell within the specified period.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Spectral-efficiency-based connected mode MLB	**INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch.*MlbAlgoSwitch*** parameter selected, and **NRCellMlb.*MlbTriggerMode*** set to **SPEC_EFF_BASED_USER_NUM**	*Mobility Load Balancing in Connected Mode*	Assume that both network slice admission control and spectral-efficiency-based connected mode MLB are enabled and UEs in a network slice are selected for MLB. If the maximum number of UEs in a target cell has been reached, the network slice UEs cannot be handed over from the source cell to the target cell. To reduce possibility of a decrease in the handover success rate due to invalid handovers, the **gNBMobilityCommParam.*ResNotAvailForSliceTimer*** parameter must be set to a value greater than 20s. In this case, all UEs in the network slice will not be handed over to the target cell within the specified period.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Radio-resource-based connected mode MLB	**INTER_FREQ_CONNECTED_MLB_SW** option of the **NRCellAlgoSwitch.*MlbAlgoSwitch*** parameter selected, and **NRCellMlb.*MlbTriggerMode*** set to **RADIO_RESOURCE**	*Mobility Load Balancing in Connected Mode*	Assume that both network slice admission control and radio-resource-based connected mode MLB are enabled and UEs in a network slice are selected for MLB. If the maximum number of UEs in a target cell has been reached, the network slice UEs cannot be handed over from the source cell to the target cell. To reduce possibility of a decrease in the handover success rate due to invalid handovers, the **gNBMobilityCommParam.*ResNotAvailForSliceTimer*** parameter must be set to a value greater than 20s. In this case, all UEs in the network slice will not be handed over to the target cell within the specified period.

NR DU cell resource management between network slices:

- Functions related to interference reduction:

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Coordinated interference management	**INTRA_GNB_DL_CS_CBF_SW** and **INTER_GNB_DL_CS_CBF_SW** options of the **NRDUCellAlgoSwitch.*CompSwitch*** parameter	*Coordinated Interference Management (Low-Frequency TDD)*	If (1) both "NR DU cell resource management between network slices" and "coordinated interference management" are enabled, and (2) the start position (only RB0 is supported in the current version) of the minimum RB quota is configured for the cooperating cell, then RB0 may not be the start position of the minimum RB quota for scheduling network slice UEs in the cooperating cell.

- Functions related to MIMO:

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDSCH MU-MIMO	**DL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch.*MuMimoSwitch*** parameter	*MIMO (TDD)*	PDSCH or PUSCH MU-MIMO works with NR DU cell resource management between network slices as follows: ● If **NRDUCellRes.*ResStrategyType*** is set to **TTI_ONLY** for a slice group, UEs in the group do not participate in pairing. ● If **NRDUCellRes.*ResStrategyType*** is set to **MEAN_ONLY** for a slice group, UEs in the group can participate in pairing. The following descriptions apply when pairing is performed: ● The proportion of RBs used by UEs in a slice group equals the proportion of paired layers for these UEs. For example, if there are a total of six paired layers and four of them are used by UEs in a slice group, then UEs in the group use 4/6 of all available RBs. ● Assume that **NRDUCellRes.*DlRbAverageRatio*** and
Low-frequency TDD	PUSCH MU-MIMO	**UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch.*MuMimoSwitch*** parameter		

RAT	Function Name	Function Switch	Reference	Description
				NRDUCellRes.*UlRbAverageRatio* are configured. The proportion of RBs used by UEs in a slice group using the mean-type control policy can reach the configured proportion only when a sufficient number of UEs in the group perform full buffer services. In this case, the pairing success rate decreases for the UEs that are outside the slice group and highly correlated with UEs in the group, leading to a decline in both average downlink cell throughput and average downlink UE throughput. This will negatively impact user experience.
High-frequency TDD	mmWave PUSCH MU-MIMO	**UL_MU_MIMO_SW** option of the **NRDUCellAlgoSwitch.*MuMimoSwitch*** parameter	*MIMO (TDD)*	When "NR DU cell resource management between network slices" and "mmWave PUSCH MU-MIMO" are both enabled, UEs in a network slice group cannot participate in pairing.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	CCE partial symbol adaptation	**CCE_PARTIAL_SYM_ADAPT_SW** option of the **NRDUCellPdcch**.***PdcchAlgoSwitch*** parameter	*Massive MIMO iBeam (Low-Frequency TDD)*	When NR DU cell resource management between network slices takes effect, the benefits of CCE partial symbol adaptation will decrease.
Low-frequency TDD	Inter-cell interference avoidance	**INTER_CELL_INTRF_AVOID_SW** option of the **NRDUCellCollabServ**.***MultiCellMimoSwitch*** parameter	*Massive MIMO iBeam (Low-Frequency TDD)*	The probability of "inter-cell interference avoidance" taking effect decreases when (1) it is enabled together with "NR DU cell resource management between network slices" and (2) the range of available RBs varies among cells.
Low-frequency TDD	Uplink 4-layer transmission for a single FWA user	**UL_SU_FOUR_LAYER_SW** option of the **NRDUCellUlRank**.***UlRankAlgoSw*** parameter	*FWA*	If uplink 4-layer transmission for a single FWA user has taken effect, whether uplink scheduling on a per RB basis takes effect for the user is affected.

- Functions related to energy saving:

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RF channel intelligent shutdown	**RF_SHUTDOWN_SW** option of the **NRDUCellAlgoSwitch.*PowerSavingSwitch*** parameter and **BANDWIDTH_BASED_POWER_AGG_SW** option of the **NRDUCellChnPwr.*ChnPwrAlgoSwitch*** parameter	*Energy Conservation and Emission Reduction*	When both "RF channel intelligent shutdown" and "NR DU cell resource management between network slices" are enabled, the former does not take effect in the scenarios described in the table "RF channel intelligent shutdown scenario for certain NR TDD low-frequency AAUs/RRUs in automatic coverage compensation mode" of *Energy Conservation and Emission Reduction*.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Power saving BWP	**BWP2_SWITCH** option of the **NRDUCellUePwrSaving.*BwpPwrSavingSw*** parameter	*UE Power Saving*	When both "NR DU cell resource management between network slices" and "power saving BWP" are enabled: ● If the TTI-based control policy is configured but **NRDUCellRes.*DlMinRbStartPos*** and **NRDUCellRes.*UlMinRbStartPos*** are not specified for the former function, power saving BWP does not take effect for all UEs in the network slice group in an NR DU cell. ● If the TTI-based control policy is configured and **NRDUCellRes.*DlMinRbStartPos*** or **NRDUCellRes.*UlMinRbStartPos*** is specified for the former function, power saving BWP does not take effect for all UEs in an NR DU cell, regardless of whether these UEs are in a network slice group.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	DRX	**BASIC_DRX_SW** option of the **NRDUCellUePwrSaving**.***NrDuCellDrxAlgoSwitch*** parameter	*DRX*	If (1) both DRX and NR DU cell resource management between network slices are enabled, and (2) the latter function is enabled for a network slice group with the TTI-based control policy, then all UEs served by the network slice group with the TTI-based control policy in an NR DU cell will not enter the DRX state, except when reportCGI measurements are performed.

- Functions related to coordination:

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UL and DL Decoupling	**NRDUCellAlgoSwitch.*UlDlDecouplingSwitch*** set to **ON**	*UL and DL Decoupling*	If (1) both "NR DU cell resource management between network slices" and "UL and DL Decoupling" are enabled, and (2) the **RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch.*NetworkSliceAlgoSwitch*** parameter is selected for an SUL cell, then this option must also be selected for the corresponding TDD cell. In this case, the percentages of RBs configured for the TDD cell apply to the SUL cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Independent SUL deployment	**SUPER_UL_TDM_SCH_SW** option of the **NRDUCellAlgoSwitch.*SuperUplinkSwitch*** parameter	*Super Uplink*	If (1) both "independent SUL deployment" and "NR DU cell resource management between network slices" are enabled, and (2) the **RB_DYNAMIC_CONTROL_SW** option of the **NRDUCellAlgoSwitch.*NetworkSliceAlgoSwitch*** parameter is selected for an SUL cell, then this option must also be selected for the corresponding TDD cell. In this case, the percentages of RBs configured for the TDD cell apply to the SUL cell.

- Functions related to RAN services:

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink RB reservation	**UL_RB_RSV_SW** option of the **NRCellAlgoSwitch.*VonrSwitch*** parameter	*VoNR*	When both "NR DU cell resource management between network slices" and "uplink RB reservation" are enabled, "uplink RB reservation" does not take effect for the VoNR UEs in the network slice group.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Uplink MU-MIMO optimization for FWA private line	**PRIVATE_LINE_UL_MUMIMO_OPT_SW** option of the **NRDUCellAlgoSwitch.*FwaAlgoSwitch*** parameter	*FWA*	After uplink MU-MIMO optimization for FWA private line is enabled, if **NRDUCellRes.*ResStrategyType*** is set to **MEAN_ONLY** for NR DU cell resource management between network slices, there is a possibility that resources are limited, decreasing the data rate of FWA private line users.
Low-frequency TDD	Downlink MU-MIMO optimization for FWA private line	**PRIVATE_LINE_DL_MUMIMO_OPT_SW** option of the **NRDUCellAlgoSwitch.*FwaAlgoSwitch*** parameter	*FWA*	After downlink MU-MIMO optimization for FWA private line is enabled, if **NRDUCellRes.*ResStrategyType*** is set to **MEAN_ONLY** for NR DU cell resource management between network slices, there is a possibility that resources are limited, decreasing the user-perceived rate of FWA private line users.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink MU-MIMO QoS guarantee	**DL_MU_MIMO_QOS_SW** option of the **NRDUCellAlgoSwitch.*FwaAlgoSwitch*** parameter	*FWA*	After downlink MU-MIMO QoS guarantee is enabled, if **NRDUCellRes.*ResStrategyType*** is set to **MEAN_ONLY** for NR DU cell resource management between network slices, there is a possibility that resources are limited, decreasing the gains yielded by downlink MU-MIMO QoS guarantee.
Low-frequency TDD	Downlink resource isolation	**DL_EMBB_PREFERENTIAL_SCH_SW** option of the **NRDUCellAlgoSwitch**.***FwaAlgoSwitch*** parameter	*FWA*	If (1) downlink resource isolation is enabled, and (2) the minimum percentage of downlink RBs (indicated by **NRDUCellRes.*DlRbMinRatio***) in the "NR DU cell resource management between network slices" function is used for FWA UEs, then downlink resource isolation will not take effect for these FWA UEs. If this minimum percentage of downlink RBs is used for non-FWA UEs under the same conditions, the reduction in average downlink throughput of FWA UEs will be amplified.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RedCap Introduction	**REDCAP_SW** option of the **NRDUCellFeatureSw**.***UeCapbBasedFunSw*** parameter	*RedCap*	If the uplink slice RB resources (indicated by a product of **NRDUCellRes.*UlRbMaxRatio*** or **NRDUCellRes.*UlRbMinRatio*** and the cell bandwidth) are greater than 15 MHz, slice UEs may consume excessive uplink scheduling resources. As a result, RedCap UEs outside the slice group may fail to be scheduled in a timely manner due to insufficient uplink scheduling resources. If the downlink slice RB resources (indicated by a product of **NRDUCellRes.*DlRbMaxRatio*** or **NRDUCellRes.*DlRbMinRatio*** and the cell bandwidth) are greater than 15 MHz, slice UEs may consume excessive downlink scheduling resources. As a result, RedCap UEs outside the slice group may fail to be scheduled in a timely manner due to insufficient downlink scheduling resources.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Flexible experience differentiation	**FLEX_EXP_DIFF_SW** option of the **NRDUCellFeatureSw**.***QciBasedDiffServExpSw*** parameter	*Service-differentiated Experience Guarantee*	After both "NR DU cell resource management between network slices" and "flexible experience differentiation" are enabled, the base station estimates the uplink or downlink peak data rate based on the available resource range of the slice to which the UE belongs and performs further data rate control. As a result, the difference between the estimated data rate and the actually measured data rate may increase.
Low-frequency TDD	Enhanced L4S congestion control based on rate adjustment	• Enhanced uplink L4S congestion control based on rate adjustment: **UL_L4SPLUS_FOR_FAIR_SW** option of the **NRDUCellUlSch**.***UlSchAlgoSwitch*** parameter • Enhanced downlink L4S congestion control based on rate adjustment: **DL_L4SPLUS_FOR_FAIR_SW** option of the **NRDUCellDlSch**.***DlSchAlgoEnhSwitch*** parameter	*Adaptive Managed Latency*	If both "NR DU cell resource management between network slices" and "enhanced L4S congestion control based on rate adjustment" are enabled, the gNodeB restricts the available RBs for L4S UEs based on the available resource range of the network slice group to which the UEs belong, which may cause the calculated maximum rate and fair rate to be inaccurate.

- Basic functions:

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink interference randomization-based scheduling	**DL_INTRF_RANDOM_SW** option of the **NRDUCellAlgoSwitch.*AdaptiveEdgeExpEnhSwitch*** parameter	*Downlink Scheduling*	If (1) both "downlink interference randomization-based scheduling" and "NR DU cell resource management between network slices" are enabled, and (2) both the minimum RB quota and its start position are configured, then "downlink interference randomization-based scheduling" is not performed within the minimum RB quota.
Low-frequency TDD	Uplink interference randomization-based scheduling	**NRDUCellAlgoSwitch.*UlIntrfRandomSchSwitch*** set to **ON**	*Uplink Scheduling*	If (1) both "uplink interference randomization-based scheduling" and "NR DU cell resource management between network slices" are enabled, and (2) both the minimum RB quota and its start position are configured, then "uplink interference randomization-based scheduling" is not performed within the minimum RB quota.

Downlink type 1 scheduling for slice groups with the TTI-based control policy:

- Functions related to interference reduction:

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Coordinated interference management	**INTRA_GNB_DL_CS_CBF_SW** and **INTER_GNB_DL_CS_CBF_SW** options of the **NRDUCellAlgoSwitch.*CompSwitch*** parameter	*Coordinated Interference Management (Low-Frequency TDD)*	When downlink type 1 scheduling for slice groups with the TTI-based control policy takes effect, coordinated interference management does not take effect for UEs in the slice groups.

- Functions related to MIMO:

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Co-MM	**INTRA_GNB_SU_COMM_SW** option of the **NRDUCellCollabServ.*MultiCellMimoSwitch*** parameter	*Co-MM (Low-Frequency TDD)*	When downlink type 1 scheduling for slice groups with the TTI-based control policy takes effect, Co-MM does not take effect for UEs in the slice groups.

- Functions related to coordination:

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Intra-base-station DL CoMP	**INTRA_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch.*CompSwitch*** parameter	*CoMP*	When downlink type 1 scheduling for slice groups with the TTI-based control policy takes effect, intra-base-station DL CoMP does not take effect for UEs in the slice groups.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Inter-base-station downlink joint transmission	**INTER_GNB_DL_JT_SW** option of the **NRDUCellAlgoSwitch.*CompSwitch*** parameter	None	When downlink type 1 scheduling for slice groups with the TTI-based control policy takes effect, inter-base-station downlink joint transmission does not take effect for UEs in the slice groups.

Single-UE multi-slice RB management policy:

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	User-rate-based dynamic network slice resource management	**DYNAMIC_NETWORKSLICE_SW** option of the **NRDUCellCommonSw**.***ServiceDiffAlgoExtSwitch*** parameter	*Network Slicing*	When both "user-rate-based dynamic network slice resource management" and "single-UE multi-slice RB management policy" take effect, the average percentage of RBs estimated by the gNodeB may be inaccurate.

Resource sharing between network slice groups and RFSP groups:

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	User-rate-based dynamic network slice resource management	**DYNAMIC_NETWORKSLICE_SW** option of the **NRDUCellCommonSw**.***ServiceDiffAlgoExtSwitch*** parameter	*Network Slicing*	When both "user-rate-based dynamic network slice resource management" and "resource sharing between network slice groups and RFSP groups" take effect, the average percentage of RBs estimated by the gNodeB for UEs in a network slice group may be inaccurate.

User-rate-based dynamic network slice resource management:

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Resource sharing between network slice groups and RFSP groups	**NRDUCellRes**.***ResUsageObject*** set to **NS_RFSP_GROUP**	*Network Slicing*	When both "user-rate-based dynamic network slice resource management" and "resource sharing between network slice groups and RFSP groups" take effect, the gNodeB may make inaccurate dynamic resource adjustments.
Low-frequency TDD	Single-UE multi-slice RB management policy	**SINGLEUE_MULTISLICE_RB_CTRL_SW** option of the **NRDUCellAlgoSwitch.*NetworkSliceAlgoSwitch*** parameter	*Network Slicing*	When both "user-rate-based dynamic network slice resource management" and "single-UE multi-slice RB management policy" take effect, the gNodeB may make inaccurate dynamic resource adjustments.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	UL and DL Decoupling	**NRDUCellAlgoSwitch.*UlDlDecouplingSwitch*** set to **ON**	*UL and DL Decoupling*	When the user-rate-based dynamic network slice resource management function is enabled for an SUL cell with UL and DL Decoupling in effect, you are advised to also enable this function for the corresponding TDD cell. Otherwise, the gNodeB may perform inaccurate dynamic resource adjustments. When UL and DL Decoupling and user-rate-based dynamic network slice resource management are enabled for both the SUL cell and the corresponding TDD cell, the values of **NRDUCellQciBearer**.***DynResUlTargetRate*** and **NRDUCellQciBearer**.***DynResDlTargetRate*** of the TDD cell also apply to the SUL cell.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Independent SUL deployment	**SUPER_UL_TDM_SCH_SW** option of the **NRDUCellAlgoSwitch.*SuperUplinkSwitch*** parameter	*Super Uplink*	When the user-rate-based dynamic network slice resource management function is enabled for an SUL cell with independent SUL deployment in effect, you are advised to also enable this function for the corresponding TDD cell. Otherwise, the gNodeB may perform inaccurate dynamic resource adjustments. When independent SUL deployment and user-rate-based dynamic network slice resource management are enabled for both the SUL cell and the corresponding TDD cell, the values of **NRDUCellQciBearer**.***DynResUlTargetRate*** and **NRDUCellQciBearer**.***DynResDlTargetRate*** of the TDD cell also apply to the SUL cell.
Low-frequency TDD	Downlink intra-band CA	**INTRA_BAND_CA_SW** option of the **NRDUCellAlgoSwitch.*CaAlgoSwitch*** parameter	*Carrier Aggregation*	When user-rate-based dynamic network slice resource management is enabled, the **NRDUCellQciBearer**.***DynResDlTargetRate*** parameter value configured for the PCell also applies to the SCells.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_CA_SW** option of the **NRDUCellAlgoSwitch.*CaAlgoSwitch*** parameter	*Carrier Aggregation*	When user-rate-based dynamic network slice resource management is enabled, the **NRDUCellQciBearer**.***DynResDlTargetRate*** parameter value configured for the PCell also applies to the SCells.
Low-frequency TDD	Uplink intra-band CA	**INTRA_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When user-rate-based dynamic network slice resource management is enabled, the **NRDUCellQciBearer**.***DynResUlTargetRate*** parameter value configured for the PCell also applies to the SCells.
Low-frequency TDD	Uplink intra-FR inter-band CA	**INTRA_FR_INTER_BAND_UL_CA_SW** option of the **NRDUCellAlgoSwitch**.***CaAlgoSwitch*** parameter	*Carrier Aggregation*	When user-rate-based dynamic network slice resource management is enabled, the **NRDUCellQciBearer**.***DynResUlTargetRate*** parameter value configured for the PCell also applies to the SCells.

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

In low frequency bands:

- For user-rate-based dynamic network slice resource management, base station models must be 3900 and 5900 series base stations (macro base stations), and 3900 series base stations must be configured with the BBU3910.
- For other functions, the requirements for base station models are as follows:

– 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
– DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

In high frequency bands, 5900 series base stations (macro base stations) are required.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

Only the UMPTg series and later boards support up to 32 network slice resource groups, while other main control boards support up to 6 network slice resource groups.

RF Modules

All NR-capable RF modules support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

4.3.4 Others

- The core network must support network slicing.
- End-to-end network slices must be deployed for UEs and the transport network.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-6 describes the parameters used for function activation. This section does not describe parameters related to cell establishment.

Table 4-6 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Operator ID	**gNBNetworkSlice.*OperatorId***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Slice Service Type	**gNBNetworkSlice.*SliceServiceType***	Set this parameter to a 3GPP-defined SST or a customized SST based on the network plan. Ensure that the configuration is identical to that on the CN.
Slice Differentiator	**gNBNetworkSlice.*SliceDifferentiator***	Set this parameter based on the network plan.
Slice Description	**gNBNetworkSlice.*SliceDescription***	Set this parameter based on the network plan.
Tracking Area ID	**gNBTaNs.*TrackingAreaId***	Set this parameter based on the network plan.
Operator ID	**gNBTaNs.*OperatorId***	Set this parameter to the same value as **gNBNetworkSlice.*OperatorId***.
Slice Service Type	**gNBTaNs.*SliceServiceType***	Set this parameter to the same value as **gNBNetworkSlice.*SliceServiceType***.
Slice Differentiator	**gNBTaNs.*SliceDifferentiator***	Set this parameter to the same value as **gNBNetworkSlice.*SliceDifferentiator***.
CU NG Type	**gNBCUNg.*CuNgType***	Set this parameter based on the network plan. Set this parameter to **DEFAULT_CP** or **DEFAULT_CP_UP** when configuring a default AMF. Set this parameter to **DEFAULT_UP** or **DEFAULT_CP_UP** when configuring a default UPF. When the control plane and user plane are configured on two NG interfaces respectively, set this parameter to **DEFAULT_CP** for one NG interface and **DEFAULT_UP** for the other. When the control plane and user plane are configured on one NG interface, set this parameter to **DEFAULT_CP_UP**.

Parameter Name	Parameter ID	Setting Notes
gNodeB CU NG ID	**gNBCUNgNs.*gNBCuNgId***	Set this parameter based on the network plan.
Operator ID	**gNBCUNgNs.*OperatorId***	Set this parameter to the same value as **gNBNetworkSlice.*OperatorId***.
Slice Service Type	**gNBCUNgNs.*SliceServiceType***	Set this parameter to the same value as **gNBNetworkSlice.*SliceServiceType***.
Slice Differentiator	**gNBCUNgNs.*SliceDifferentiator***	Set this parameter to the same value as **gNBNetworkSlice.*SliceDifferentiator***.
Network Slice Algorithm Switch	**NRCellAlgoSwitch.*NetworkSliceAlgoSwitch***	To enable admission control based on network slices, select the **ADMISSION_BASED_ON_NS_ID_SW** option of this parameter.
Network Slice Algorithm Switch	**NRCellAlgoSwitch.*NetworkSliceAlgoSwitch***	To enable network slice resource management, select the **ENHANCED_NETWORKSLICE_SW** option of this parameter.
NR Cell ID	**NRCellRes.*NrCellId***	Set this parameter based on the network plan.
Cell Resource ID	**NRCellRes.*CellResId***	Set this parameter based on the network plan.
Max RRC Connected User Number	**NRCellRes.*MaxRrcConnUserNum***	Set this parameter based on the network plan.
Slice not Supported Timer	**gNBMobilityCommParam.*SliceNotSupportedTimer***	Set this parameter based on the network plan.
Resource not Available for Slice Timer	**gNBMobilityCommParam.*ResNotAvailForSliceTimer***	Set this parameter based on the network plan.
NR Cell ID	**NRCellNsGrp.*NrCellId***	Set this parameter to the same value as **NRCellRes.*NrCellId***.

Parameter Name	Parameter ID	Setting Notes
Network Slice Group ID	**NRCellNsGrp.***NsGrpId*	Set this parameter based on the network plan.
Cell Resource ID	**NRCellNsGrp.***CellResId*	Set this parameter to the same value as **NRCellRes.***CellResId*.
Network Slice Group Description	**NRCellNsGrp.***NsGrpDescription*	Set this parameter based on the network plan.
NR Cell ID	**NRCellNsGrpNs.***NrCellId*	Set this parameter to the same value as **NRCellRes.***NrCellId*.
Network Slice Group ID	**NRCellNsGrpNs.***NsGrpId*	Set this parameter to the same value as **NRCellNsGrp.***NsGrpId*.
Operator ID	**NRCellNsGrpNs.***OperatorId*	Set this parameter to the same value as **gNBNetworkSlice.***OperatorId*.
Slice Service Type	**NRCellNsGrpNs.***SliceServiceType*	Set this parameter to the same value as **gNBNetworkSlice.***SliceServiceType*.
Slice Differentiator	**NRCellNsGrpNs.***SliceDifferentiator*	Set this parameter to the same value as **gNBNetworkSlice.***SliceDifferentiator*.
Associated Endpoint Group Mode	**gNBCUNgNs.***AssocEpGroupMode*	Set this parameter based on the network plan. If a network slice configured on the gNodeB needs to be associated with the control-plane or user-plane endpoint groups configured for the NG interface, set this parameter to **CP** or **UP**, respectively. If a network slice configured on the gNodeB needs to be associated with both the control-plane and user-plane endpoint groups configured for the NG interface, set this parameter to **CP_UP**.

Parameter Name	Parameter ID	Setting Notes
Network Slice Algorithm Switch	**NRDUCellAlgoSwitch.*NetworkSliceAlgoSwitch***	To enable the function of NR DU cell resource management between network slices, select the **RB_DYNAMIC_CONTROL_SW** option of this parameter. The configuration of NR DU cell RB management between network slices takes effect only when this option is selected. The configuration includes network slice groups, slices in network slice groups, percentage of RBs in NR DU cell resources, start RB positions, and RB control policies.
NR DU Cell ID	**NRDUCellRes.*NrDuCellId***	Set this parameter based on the network plan.
Cell Resource ID	**NRDUCellRes.*CellResId***	Set this parameter based on the network plan.
Resource Usage Object	**NRDUCellRes.*ResUsageObject***	Set this parameter to **NETWORK_SLICE_GROUP** or **NS_RFSP_GROUP** based on the network plan. If this parameter is set to **NS_RFSP_GROUP**, the **NRDUCellRes.*ResStrategyType*** parameter can only be set to **MEAN_ONLY**.

Parameter Name	Parameter ID	Setting Notes
Resource Strategy Type	**NRDUCellRes.*ResStrategyType***	Set this parameter based on the network plan. Note the following: The TTI-based control policy is recommended only for secure isolation scenarios or for low-latency and high-reliability services. If this policy is used for a network slice group, there must be a specific maximum number of RRC_CONNECTED UEs (specified by the **NRCellRes.*MaxRrcConnUserNum*** parameter) configured for the slice group, as secure isolation scenarios or low-latency and high-reliability services rely on networking planning and cell configurations to meet the SLA requirements of UEs in the slice group. For details, contact Huawei technical support. The mean-type control policy is recommended only in scenarios other than low-latency and high-reliability scenarios. **NRDUCellRes.*DlRbAverageRatio*** or **NRDUCellRes.*UlRbAverageRatio*** can be configured to ensure resources of network slice groups that are not used for low latency and high reliability. As such, it is recommended that **NRDUCellRes.*DlRbMinRatio*** and **NRDUCellRes.*UlRbMinRatio*** not be configured when the mean-type control policy is used. Otherwise, the RB usage of the cell is affected.

Parameter Name	Parameter ID	Setting Notes
Downlink RB Max Ratio	**NRDUCellRes.*DlRbMaxRatio***	Set this parameter based on the network plan. Ensure that the number of RBs corresponding to the percentage of RBs specified by this parameter is greater than or equal to the number of RBs in one RBG.
Downlink RB Average Ratio	**NRDUCellRes.*DlRbAverageRatio***	Set this parameter based on the network plan. When the TTI-based control policy is used and the **NRDUCellRes.*DlRbAverageRatio*** parameter is set to a value other than the default, that value must be the same as the value of the **NRDUCellRes.*DlRbMaxRatio*** parameter. There are no special requirements for using the mean-type control policy.
Downlink RB Min Ratio	**NRDUCellRes.*DlRbMinRatio***	Set this parameter based on the network plan.
Downlink Min RB Start Position	**NRDUCellRes.*DlMinRbStartPos***	In the current version, this parameter can only be set to **0** or left empty. If there is interference from neighboring cells, it is recommended that this parameter be set to **0** and the **INIT_BWP_FULL_BW_SW** option of the **NRDUCellAlgoSwitch.*BwpConfigPolicySwitch*** parameter be selected to ensure that scheduling of the UEs in the network slice group begins with the start position of the minimum RB quota.
Uplink RB Max Ratio	**NRDUCellRes.*UlRbMaxRatio***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Uplink RB Average Ratio	**NRDUCellRes.*UlRbAverageRatio***	Set this parameter based on the network plan. When the TTI-based control policy is used and the **NRDUCellRes.*UlRbAverageRatio*** parameter is set to a value other than the default, that value must be the same as the value of the **NRDUCellRes.*UlRbMaxRatio*** parameter. There are no special requirements for using the mean-type control policy.
Uplink RB Resource Scheduling Priority	**NRDUCellUlSch.*UlRbResSchPriority***	Set this parameter based on the network plan.
Uplink RB Min Ratio	**NRDUCellRes.*UlRbMinRatio***	Set this parameter based on the network plan.
Uplink Min RB Start Position	**NRDUCellRes.*UlMinRbStartPos***	In the current version, this parameter can only be set to **0** or left empty. If interference from neighboring cells is present, it is recommended that this parameter be set to **0**. In addition, it is recommended that the **INIT_BWP_FULL_BW_SW** option of the **NRDUCellAlgoSwitch.*BwpConfigPolicySwitch*** parameter be selected to ensure that scheduling of the UEs in the network slice group begins with the start position of the minimum RB quota.
NR DU Cell ID	**NRDUCellNsGrp.*NrDuCellId***	Set this parameter based on the network plan.
Network Slice Group ID	**NRDUCellNsGrp.*NsGrpId***	Set this parameter based on the network plan.
Cell Resource ID	**NRDUCellNsGrp.*CellResId***	Set this parameter to the same value as **NRDUCellRes.*CellResId***.
NR DU Cell ID	**NRDUCellNsGrpNs.*NrDuCellId***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Network Slice Group ID	**NRDUCellNsGrpNs.*NsGrpId***	Set this parameter to the same value as **NRDUCellNsGrp.*NsGrpId***.
Operator ID	**NRDUCellNsGrpNs.*OperatorId***	Set this parameter to the same value as **gNBNetworkSlice.*OperatorId***.
Slice Service Type	**NRDUCellNsGrpNs.*SliceServiceType***	Set this parameter to the same value as **gNBNetworkSlice.*SliceServiceType***.
Slice Differentiator	**NRDUCellNsGrpNs.*SliceDifferentiator***	Set this parameter to the same value as **gNBNetworkSlice.*SliceDifferentiator***.

Parameter Name	Parameter ID	Setting Notes
Scheduling Policy Switch	**NRDUCellRes.*SchedulingPolicySwitch***	Based on the network plan, select the **FIX_RES_UL_INTRF_RANDOM_SCH_SW** or **FIX_RES_DL_INTRF_RANDOM_SCH_SW** option of this parameter when uplink or downlink interference randomization is required. Note that the following configurations are required after this option is selected: ● Uplink interference randomization requires the following additional configurations: – **NRDUCellRes.*UlMinRbStartPos*** is set to **0**, indicating that the fixed reserved resources are allocated starting from the RB following the last RB occupied by the PUCCH. – **NRDUCellRes.*UlRbMaxRatio*** and **NRDUCellRes.*UlRbMinRatio*** have the same value. ● Downlink interference randomization requires the following additional configurations: – **NRDUCellRes.*DlMinRbStartPos*** is set to **0**, indicating that the fixed reserved resources are allocated starting from RB0. – **NRDUCellRes.*DlRbMaxRatio*** and **NRDUCellRes.*DlRbMinRatio*** have the same value. Based on the network plan, select the **DL_TYPE1_SCH_SW** option of this parameter for the slice groups with the TTI-based control policy to

Parameter Name	Parameter ID	Setting Notes
		support downlink type 1 scheduling; select the **UL_1RB_SCH_SW** option of this parameter for the slice groups with the TTI-based control policy to support uplink scheduling on a per RB basis.
Network Slice Algorithm Switch	**NRDUCellAlgoSwitch**.***NetworkSliceAlgoSwitch***	To enable the single-UE multi-slice RB management policy, select the **SINGLEUE_MULTISLICE_RB_CTRL_SW** option of this parameter. When this option is selected, the **NRDUCellRes.*DlMinRbStartPos*** parameter must be set to **65535**.
Network slice/ RFSP group sharing policy	**NRDUCellNsMgmtConfig**.***NetSliceRfspGrpSharePol***	Set this parameter based on the network plan. It is recommended that this parameter be set to **SAME_OPERATOR_PREFERRED** when both "NR DU cell resource management between network slices" and "operator-specific resource sharing enhancement" need to take effect.
Network slice/ RFSP group preemption policy	**NRDUCellNsMgmtConfig.*NetSliceRfspGrpPreemptPol***	Set this parameter based on the network plan.
Service Differentiation Algorithm Ext Sw	**NRDUCellCommonSw**.***ServiceDiffAlgoExtSwitch***	Select the **DYNAMIC_NETWORKSLICE_SW** option of this parameter when user-rate-based dynamic network slice resource management is required.
Dynamic Resource Uplink Target Rate	**NRDUCellQciBearer**.***DynResUlTargetRate***	Set this parameter based on the network plan.
Dynamic Resource Downlink Target Rate	**NRDUCellQciBearer**.***DynResDlTargetRate***	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
MAC Parameter Group ID	**NRDUCellQciBearer**.***MacParamGroupId***	Configure extended QCIs based on the network plan. Standardized QCIs that have been created by default do not need to be configured.

4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

NOTICE

- If (1) an operator has multiple NG interfaces (configured by the **gNBCUNg** MO), and (2) these NG interfaces use the same user-plane host IP address of a user-plane endpoint group and the same transmission policy (meaning the same VLAN ID on the transport network), then these NG interfaces must be associated with the same user-plane endpoint group (configured by **gNBCUNg.*UpEpGroupId***).
- Assume that RBG-granularity scheduling is used. When configuring NR DU cell resources for a network slice, ensure that both the maximum number of available RBs per network slice group and those for outside network slice groups are not less than the number of RBs in one RBG. That is, all of the following conditions must be met:
  - Number of RBs corresponding to the percentage of RBs specified by **NRDUCellRes.*DlRbMaxRatio*** or **NRDUCellRes.*UlRbMaxRatio*** ≥ Number of RBs in one RBG
  - Total number of available PDSCH RBs in a cell – Sum of RBs corresponding to the percentage of RBs specified by **NRDUCellRes.*DlRbMinRatio*** or **NRDUCellRes.*DlRbAverageRatio*** for all network slice groups ≥ Number of RBs in one RBG. If **NRDUCellRes.*DlRbAverageRatio*** is configured for a network slice group, **NRDUCellRes.*DlRbAverageRatio*** is applied in this formula. Otherwise, **NRDUCellRes.*DlRbMinRatio*** is applied.
  - Total number of available PUSCH RBs in a cell – Sum of RBs corresponding to the percentage of RBs specified by **NRDUCellRes.*UlRbMinRatio*** or **NRDUCellRes.*UlRbAverageRatio*** for all network slice groups ≥ Number of RBs in one RBG. If **NRDUCellRes.*UlRbAverageRatio*** is configured for a network slice group, **NRDUCellRes.*UlRbAverageRatio*** is applied in this formula. Otherwise, **NRDUCellRes.*UlRbMinRatio*** is applied.

  For details about the requirements for configuring the parameters related to RB percentages, see **Table 4-3**.

Activation Command Examples (Single Operator and Single Network Slice)

Deploying a Network Slice

- Configuring a network slice on the gNodeB
  ```
  ADD GNBNETWORKSLICE: OperatorId=0, SliceServiceType=1, SliceDifferentiator=6543, SliceDescription="EMBB";
  ```
- Configuring a default AMF (It can be the one connected over an existing NG interface or over a new NG interface created for the network slice, depending on the network plan.)
  - Example based on the assumption that the AMF connected over an existing NG interface is used
    ```
    MOD GNBCUNG: gNBCuNgId=0, CuNgType=DEFAULT_CP_UP;
    ```
  - Example based on the assumption that the AMF connected over a new NG interface created for the network slice is used
    ```
    ADD GNBCUNG: gNBCuNgId=2, CpEpGroupId=2, UpEpGroupId=2, CuNgType=DEFAULT_CP_UP;
    ```
- Configuring a network slice specific to a tracking area on the gNodeB
  ```
  ADD GNBTANS: TrackingAreaId=0, OperatorId=0, SliceServiceType=1, SliceDifferentiator=6543;
  ```
- Configuring the network slice for a gNodeB CU NG interface (the NG interface associated with the default AMF, or a new NG interface created for

the network slice but not associated with the default AMF, depending on the network plan)

– (Required if the NG interface is the one that is associated with the default AMF) Directly binding the NG interface

```
ADD GNBCUNGNS: gNBCuNgId=0, OperatorId=0, SliceServiceType=1, SliceDifferentiator=6543,
AssocEpGroupMode=CP_UP;
```

– (Required if the NG interface is the one that is created for the network slice but not associated with the default AMF) Configuring and binding the NG interface

```
ADD GNBCUNG: gNBCuNgId=3, CpEpGroupId=3, UpEpGroupId=3, CuNgType=NORMAL;
ADD GNBCUNGNS: gNBCuNgId=3, OperatorId=0, SliceServiceType=1, SliceDifferentiator=6543;
```

Configuring Network Slice Admission Control

```
//Enabling admission control based on network slice identifiers
MOD NRCELLALGOSWITCH: NrCellId=7, NetworkSliceAlgoSwitch=ADMISSION_BASED_ON_NS_ID_SW-1;
//Configuring the slice-not-supported timer and slice-resource-unavailable timer
MOD GNBMOBILITYCOMMPARAM: SliceNotSupportedTimer=1, ResNotAvailForSliceTimer=10;
```

Configuring Network Slice Resource Management

```
//Enabling network slicing enhancement
MOD NRCELLALGOSWITCH: NrCellId=7, NetworkSliceAlgoSwitch=ENHANCED_NETWORKSLICE_SW-1;
```

Configuring NR Cell Resources for the Network Slice

```
//Configuring NR cell resources
ADD NRCELLRES: NrCellId=7, CellResId=0, MaxRrcConnUserNum=20;
//Configuring a network slice group for the NR cell and binding the NR cell resources to the group
ADD NRCELLNSGRP: NrCellId=7, NsGrpId=0, CellResId=0, NsGrpDescription="GRP_CU";
//Configuring a network slice in the network slice group of the NR cell
ADD NRCELLNSGRPNS: NrCellId=7, NsGrpId=0, OperatorId=0, SliceServiceType=1, SliceDifferentiator=6543;
```

Configuring the Shared Resources Between the Network Slice Group and RFSP Group in an NR Cell (For details about RFSP groups, see *Resource Control in NSA Networking*.)

```
//Configuring NR cell resources
ADD NRCELLRES: NrCellId=7, CellResId=0, MaxRrcConnUserNum=10;
//Configuring a network slice group and an RFSP group for the NR cell and binding the NR cell resources to
the groups
ADD NRCELLNSGRP: NrCellId=7, NsGrpId=0, CellResId=0, NsGrpDescription="GRP_CU";
ADD NRCELLOPRFSP: NrCellId=7, OperatorId=0, RfspIndex=1, RfspGrpCellResId=1;
//Configuring a network slice in the network slice group of the NR cell
ADD NRCELLNSGRPNS: NrCellId=7, NsGrpId=0, OperatorId=0, SliceServiceType=1, SliceDifferentiator=6543;
```

Configuring NR DU Cell Resources for the Network Slice

```
//Enabling dynamic RB control
MOD NRDUCELLALGOSWITCH: NrDuCellId=7, NetworkSliceAlgoSwitch=RB_DYNAMIC_CONTROL_SW-1;
//Configuring NR DU cell resources
ADD NRDUCELLRES: NrDuCellId=7, CellResId=0, ResUsageObject=NETWORK_SLICE_GROUP,
ResStrategyType=MEAN_ONLY, DlRbMaxRatio=30, DlRbAverageRatio=30, DlRbMinRatio=20,
UlRbMaxRatio=30, UlRbAverageRatio=30, UlRbMinRatio=20, DlMinRbStartPos=65535,
UlMinRbStartPos=65535;
//Configuring a network slice group for the NR DU cell and binding the NR DU cell resources to the group
ADD NRDUCELLNSGRP: NrDuCellId=7, NsGrpId=0, CellResId=0, NsGrpDescription="GRP_DU";
//Configuring a network slice in the network slice group of the NR DU cell
ADD NRDUCELLNSGRPNS: NrDuCellId=7, NsGrpId=0, OperatorId=0, SliceServiceType=1,
SliceDifferentiator=6543;
//Configuring the uplink scheduling priority policy for UEs in a network slice group in an NR DU cell
MOD NRDUCELLULSCH: NrDuCellId=7, UlRbResSchPriority=NOT_CONFIG;
```

Configuring the Shared Resources Between the Network Slice Group and RFSP Group in an NR DU Cell (For details about RFSP groups, see *Resource Control in NSA Networking*.)

```
//Enabling dynamic RB control
MOD NRDUCELLALGOSWITCH: NrDuCellId=7, NetworkSliceAlgoSwitch=RB_DYNAMIC_CONTROL_SW-1;
MOD NRDUCELLCOMMONSW: NrDuCellId=7, RfspAlgoSwitch=RB_DYNAMIC_CONTROL_SW-1;
//Configuring NR DU cell resources
ADD NRDUCELLRES: NrDuCellId=7, CellResId=0, ResUsageObject=NS_RFSP_GROUP, DlRbMaxRatio=10,
DlRbMinRatio=10, DlRbAverageRatio=10, UlRbMaxRatio=10, UlRbMinRatio=10, UlRbAverageRatio=10;
//Configuring a network slice group and an RFSP group for the NR DU cell and binding the NR DU cell
resources to the groups
ADD NRDUCELLNSGRP: NrDuCellId=7, NsGrpId=0, CellResId=0, NsGrpDescription="GRP_DU";
ADD NRDUCELLOPRFSP: NrDuCellId=7, OperatorId=0, RfspIndex=1, RfspGrpCellResId=1;
//Configuring a network slice in the network slice group of the NR DU cell
ADD NRDUCELLNSGRPNS: NrDuCellId=7, NsGrpId=0, OperatorId=0, SliceServiceType=1,
SliceDifferentiator=6543;
```

Enabling User-Rate-based Dynamic Network Slice Resource Management

```
//Configuring the ID of the MAC parameter group referenced by a bearer in an NR DU cell, with the QCI
specified for UEs in a network slice group (example uplink/downlink target rate: 10000)
MOD NRDUCELLQCIBEARER: NrDuCellId=7, Qci=128, MacParamGroupId=9, DynResUlTargetRate=10000,
DynResDlTargetRate=10000;
//Turning on the switch for user-rate-based dynamic network slice resource management
MOD NRDUCELLCOMMONSW: NrDuCellId=7, ServiceDiffAlgoExtSwitch=DYNAMIC_NETWORKSLICE_SW-1;
```

Activation Command Examples (Single Operator and Multiple Network Slices)

Deploying Network Slices (assuming that three network slices are deployed for a single operator)

```
//Configuring network slices on the gNodeB
ADD GNBNETWORKSLICE: OperatorId=0, SliceServiceType=1, SliceDifferentiator=0,
SliceDescription="EMBB_1";
ADD GNBNETWORKSLICE: OperatorId=0, SliceServiceType=1, SliceDifferentiator=1,
SliceDescription="EMBB_2";
ADD GNBNETWORKSLICE: OperatorId=0, SliceServiceType=2, SliceDifferentiator=6543,
SliceDescription="URLLC";
//Configuring a default AMF (In single-operator scenarios, at least one default AMF must be configured.
The following example assumes that the one connected over an existing NG interface is used.)
MOD GNBCUNG: gNBCuNgId=0, CuNgType=DEFAULT_CP_UP;
//Configuring the network slices for a tracking area on the gNodeB
ADD GNBTANS: TrackingAreaId=0, OperatorId=0, SliceServiceType=1, SliceDifferentiator=0;
ADD GNBTANS: TrackingAreaId=0, OperatorId=0, SliceServiceType=1, SliceDifferentiator=1;
ADD GNBTANS: TrackingAreaId=0, OperatorId=0, SliceServiceType=2, SliceDifferentiator=6543;
//Configuring the network slices for a gNodeB CU NG interface (Each network slice must be associated with
an NG interface, and different network slices can be associated with the same or different NG interfaces. In
the following examples, multiple network slices are associated with the same NG interface.)
ADD GNBCUNGNS: gNBCuNgId=0, OperatorId=0, SliceServiceType=1, SliceDifferentiator=0;
ADD GNBCUNGNS: gNBCuNgId=0, OperatorId=0, SliceServiceType=1, SliceDifferentiator=1;
ADD GNBCUNGNS: gNBCuNgId=0, OperatorId=0, SliceServiceType=2, SliceDifferentiator=6543;
```

Configuring Network Slice Admission Control (assuming that two network slices are deployed in the same cell)

```
//Enabling admission control based on network slice identifiers
MOD NRCELLALGOSWITCH: NrCellId=7, NetworkSliceAlgoSwitch=ADMISSION_BASED_ON_NS_ID_SW-1;
```

Configuring Network Slice Resource Management

```
//Enabling network slicing enhancement
MOD NRCELLALGOSWITCH: NrCellId=7, NetworkSliceAlgoSwitch=ENHANCED_NETWORKSLICE_SW-1;
```

Configuring NR Cell Resources for Network Slices (assuming that two network slices are deployed in the same cell)

```
//Configuring NR cell resources
ADD NRCELLRES: NrCellId=7, CellResId=0, MaxRrcConnUserNum=20;
//Configuring a network slice group for the NR cell and binding the NR cell resources to the group
```



```
ADD NRCELLNSGRP: NrCellId=7, NsGrpId=0, CellResId=0, NsGrpDescription="GRP_CU";
//Configuring the network slices in the network slice group of the NR cell
ADD NRCELLNSGRPNS: NrCellId=7, NsGrpId=0, OperatorId=0, SliceServiceType=1, SliceDifferentiator=0;
ADD NRCELLNSGRPNS: NrCellId=7, NsGrpId=0, OperatorId=0, SliceServiceType=2, SliceDifferentiator=6543;
```

Configuring the Shared Resources Between the Network Slice Group and RFSP Group in an NR Cell (For details about RFSP groups, see *Resource Control in NSA Networking*.)

```
//Configuring NR cell resources
ADD NRCELLRES: NrCellId=7, CellResId=1, MaxRrcConnUserNum=10;
//Configuring a network slice group and an RFSP group for the NR cell and binding the NR cell resources to
the groups
ADD NRCELLNSGRP: NrCellId=7, NsGrpId=1, CellResId=1, NsGrpDescription="GRP_CU";
ADD NRCELLOPRFSP: NrCellId=7, OperatorId=0, RfspIndex=1, RfspGrpCellResId=1;
//Configuring a network slice in the network slice group of the NR cell
ADD NRCELLNSGRPNS: NrCellId=7, NsGrpId=1, OperatorId=0, SliceServiceType=1, SliceDifferentiator=1;
```

Configuring NR DU Cell Resources for Network Slices (assuming that two network slices are deployed in the same cell with neighboring cell interference)

```
//Configuration for the local cell
//Enabling dynamic RB control
MOD NRDUCELLALGOSWITCH: NrDuCellId=7, NetworkSliceAlgoSwitch=RB_DYNAMIC_CONTROL_SW-1;
//Configuring NR DU cell resources, the RB resource control policy, and uplink and downlink interference
randomization
ADD NRDUCELLRES: NrDuCellId=7, CellResId=0, ResUsageObject=NETWORK_SLICE_GROUP,
ResStrategyType=TTI_ONLY, DlRbMaxRatio=30, DlRbAverageRatio=30, DlRbMinRatio=30, UlRbMaxRatio=30,
UlRbAverageRatio=30, UlRbMinRatio=30, DlMinRbStartPos=0, UlMinRbStartPos=0,
SchedulingPolicySwitch=FIX_RES_UL_INTRF_RANDOM_SCH_SW-1&FIX_RES_DL_INTRF_RANDOM_SCH_SW-1
&DL_TYPE1_SCH_SW-0&UL_1RB_SCH_SW-0;
//Configuring a network slice group for the NR DU cell and binding the NR DU cell resources to the group
ADD NRDUCELLNSGRP: NrDuCellId=7, NsGrpId=0, CellResId=0, NsGrpDescription="GRP_DU";
//(Required only for URLLC network slices) Configuring a network slice in the network slice group of the NR
DU cell
ADD NRDUCELLNSGRPNS: NrDuCellId=7, NsGrpId=0, OperatorId=0, SliceServiceType=2,
SliceDifferentiator=6543;
//Configuring the uplink scheduling priority policy for UEs in a network slice group in an NR DU cell
MOD NRDUCELLULSCH: NrDuCellId=7, UlRbResSchPriority=NOT_CONFIG;
//Turning on the switch for the single-UE multi-slice RB management policy only when the minimum
downlink RB start position is set to 65535
MOD NRDUCELLALGOSWITCH: NrDuCellId=7,
NetworkSliceAlgoSwitch=SINGLEUE_MULTISLICE_RB_CTRL_SW-1;
//Configuration for the interfering neighboring cell, which is required if uplink/downlink interference
randomization has been enabled
//Enabling network slicing enhancement
MOD NRCELLALGOSWITCH: NrCellId=9, NetworkSliceAlgoSwitch=ENHANCED_NETWORKSLICE_SW-1;
//Enabling dynamic RB control
MOD NRDUCELLALGOSWITCH: NrDuCellId=9, NetworkSliceAlgoSwitch=RB_DYNAMIC_CONTROL_SW-1;
//Configuring NR DU cell resources and the RB resource control policy
ADD NRDUCELLRES: NrDuCellId=9, CellResId=0, ResUsageObject=NETWORK_SLICE_GROUP,
ResStrategyType=TTI_ONLY, DlRbMaxRatio=30, DlRbAverageRatio=30, DlRbMinRatio=30, UlRbMaxRatio=30,
UlRbAverageRatio=30, UlRbMinRatio=30, DlMinRbStartPos=0, UlMinRbStartPos=0,
SchedulingPolicySwitch=FIX_RES_UL_INTRF_RANDOM_SCH_SW-1&FIX_RES_DL_INTRF_RANDOM_SCH_SW-1;
//Configuring a network slice group for the NR DU cell and binding the NR DU cell resources to the group
ADD NRDUCELLNSGRP: NrDuCellId=9, NsGrpId=0, CellResId=0, NsGrpDescription="GRP_DU";
//(Required only for URLLC network slices) Configuring a network slice in the network slice group of the NR
DU cell
ADD NRDUCELLNSGRPNS: NrDuCellId=9, NsGrpId=0, OperatorId=0, SliceServiceType=2,
SliceDifferentiator=6543;
```

Configuring the Shared Resources Between the Network Slice Group and RFSP Group in an NR DU Cell (For details about RFSP groups, see *Resource Control in NSA Networking*.)

```
//Enabling dynamic RB control
MOD NRDUCELLALGOSWITCH: NrDuCellId=7, NetworkSliceAlgoSwitch=RB_DYNAMIC_CONTROL_SW-1;
MOD NRDUCELLCOMMONSW: NrDuCellId=7, RfspAlgoSwitch=RB_DYNAMIC_CONTROL_SW-1;
```

```
//Configuring NR DU cell resources
ADD NRDUCELLRES: NrDuCellId=7, CellResId=1, ResUsageObject=NS_RFSP_GROUP, DlRbMaxRatio=10,
DlRbMinRatio=10, DlRbAverageRatio=10, UlRbMaxRatio=10, UlRbMinRatio=10, UlRbAverageRatio=10;
//Configuring a network slice group and an RFSP group for the NR DU cell and binding the NR DU cell
resources to the groups
ADD NRDUCELLNSGRP: NrDuCellId=7, NsGrpId=1, CellResId=1, NsGrpDescription="GRP_DU";
ADD NRDUCELLOPRFSP: NrDuCellId=7, OperatorId=0, RfspIndex=1, RfspGrpCellResId=1;
//Configuring a network slice in the network slice group of the NR DU cell
ADD NRDUCELLNSGRPNS: NrDuCellId=7, NsGrpId=1, OperatorId=0, SliceServiceType=1,
SliceDifferentiator=0;
```

Enabling User-Rate-based Dynamic Network Slice Resource Management

```
//Configuring the ID of the MAC parameter group referenced by a bearer in an NR DU cell, with the QCI
specified for UEs in a network slice group (example uplink/downlink target rate: 10000)
MOD NRDUCELLQCIBEARER: NrDuCellId=7, Qci=128, MacParamGroupId=9, DynResUlTargetRate=10000,
DynResDlTargetRate=10000;
//Turning on the switch for user-rate-based dynamic network slice resource management
MOD NRDUCELLCOMMONSW: NrDuCellId=7, ServiceDiffAlgoExtSwitch=DYNAMIC_NETWORKSLICE_SW-1;
```

Activation Command Examples (Multiple Operators and Multiple Network Slices)

Deploying Network Slices (assuming that each of two operators deploys two network slices)

```
//Configuring network slices on the gNodeB
ADD GNBNETWORKSLICE: OperatorId=0, SliceServiceType=1, SliceDifferentiator=0, SliceDescription="EMBB";
ADD GNBNETWORKSLICE: OperatorId=0, SliceServiceType=2, SliceDifferentiator=6543,
SliceDescription="URLLC";
ADD GNBNETWORKSLICE: OperatorId=1, SliceServiceType=1, SliceDifferentiator=0, SliceDescription="EMBB";
ADD GNBNETWORKSLICE: OperatorId=1, SliceServiceType=2, SliceDifferentiator=6543,
SliceDescription="URLLC";
//Configuring a default AMF (In multi-operator scenarios, each operator must be configured with at least
one default AMF. The following example assumes that an operator's default AMF is that connected over an
existing NG interface.)
MOD GNBCUNG: gNBCuNgId=0, CuNgType=DEFAULT_CP_UP;
//Configuring the network slices for a tracking area on the gNodeB
ADD GNBTANS: TrackingAreaId=0, OperatorId=0, SliceServiceType=1, SliceDifferentiator=0;
ADD GNBTANS: TrackingAreaId=0, OperatorId=0, SliceServiceType=2, SliceDifferentiator=6543;
ADD GNBTANS: TrackingAreaId=0, OperatorId=1, SliceServiceType=1, SliceDifferentiator=0;
ADD GNBTANS: TrackingAreaId=0, OperatorId=1, SliceServiceType=2, SliceDifferentiator=6543;
//Configuring the network slices for a gNodeB CU NG interface (Each network slice must be associated with
an NG interface, and different network slices can be associated with the same or different NG interfaces. In
the following examples, multiple network slices are associated with the same NG interface.)
ADD GNBCUNGNS: gNBCuNgId=0, OperatorId=0, SliceServiceType=1, SliceDifferentiator=0;
ADD GNBCUNGNS: gNBCuNgId=0, OperatorId=0, SliceServiceType=2, SliceDifferentiator=6543;
ADD GNBCUNGNS: gNBCuNgId=0, OperatorId=1, SliceServiceType=1, SliceDifferentiator=0;
ADD GNBCUNGNS: gNBCuNgId=0, OperatorId=1, SliceServiceType=2, SliceDifferentiator=6543;
```

Configuring Network Slice Admission Control (assuming that network slices of two operators are deployed in the same cell)

```
//Enabling admission control based on network slice identifiers
MOD NRCELLALGOSWITCH: NrCellId=7, NetworkSliceAlgoSwitch=ADMISSION_BASED_ON_NS_ID_SW-1;
```

Configuring Network Slice Resource Management

```
//Enabling network slicing enhancement
MOD NRCELLALGOSWITCH: NrCellId=7, NetworkSliceAlgoSwitch=ENHANCED_NETWORKSLICE_SW-1;
```

Configuring NR Cell Resources for Network Slices (assuming that network slices of two operators are deployed in the same cell)

```
//Configuring NR cell resources
ADD NRCELLRES: NrCellId=7, CellResId=0, MaxRrcConnUserNum=20;
ADD NRCELLRES: NrCellId=7, CellResId=1, MaxRrcConnUserNum=20;
```



```
//Configuring network slice groups for the NR cell and binding the NR cell resources to the group
ADD NRCELLNSGRP: NrCellId=7, NsGrpId=0, CellResId=0, NsGrpDescription="GRP_CU";
ADD NRCELLNSGRP: NrCellId=7, NsGrpId=1, CellResId=1, NsGrpDescription="GRP_CU";
//Configuring the network slices in the network slice groups of the NR cell
ADD NRCELLNSGRPNS: NrCellId=7, NsGrpId=0, OperatorId=0, SliceServiceType=1, SliceDifferentiator=0;
ADD NRCELLNSGRPNS: NrCellId=7, NsGrpId=0, OperatorId=0, SliceServiceType=2, SliceDifferentiator=6543;
ADD NRCELLNSGRPNS: NrCellId=7, NsGrpId=1, OperatorId=1, SliceServiceType=2, SliceDifferentiator=6543;
```

Configuring the Shared Resources Between the Network Slice Group and RFSP Group in an NR Cell (For details about RFSP groups, see *Resource Control in NSA Networking*.)

```
//Configuring NR cell resources
ADD NRCELLRES: NrCellId=7, CellResId=2, MaxRrcConnUserNum=10;
//Configuring a network slice group and an RFSP group for the NR cell and binding the NR cell resources to
the groups
ADD NRCELLNSGRP: NrCellId=7, NsGrpId=2, CellResId=2, NsGrpDescription="GRP_CU";
ADD NRCELLOPRFSP: NrCellId=7, OperatorId=1, RfspIndex=1, RfspGrpCellResId=2;
//Configuring a network slice in the network slice group of the NR cell
ADD NRCELLNSGRPNS: NrCellId=7, NsGrpId=2, OperatorId=1, SliceServiceType=1, SliceDifferentiator=0;
```

Configuring NR DU Cell Resources for Network Slices (assuming that network slices of two operators are deployed in the same cell)

```
//Configuration for the local cell
//Enabling dynamic RB control
MOD NRDUCELLALGOSWITCH: NrDuCellId=7, NetworkSliceAlgoSwitch=RB_DYNAMIC_CONTROL_SW-1;
//Configuring NR DU cell resources, the RB resource control policy, and uplink and downlink interference
randomization
ADD NRDUCELLRES: NrDuCellId=7, CellResId=0, ResUsageObject=NETWORK_SLICE_GROUP,
ResStrategyType=TTI_ONLY, DlRbMaxRatio=30, DlRbAverageRatio=30, DlRbMinRatio=30, UlRbMaxRatio=30,
UlRbAverageRatio=30, UlRbMinRatio=30, DlMinRbStartPos=0, UlMinRbStartPos=0,
SchedulingPolicySwitch=FIX_RES_UL_INTRF_RANDOM_SCH_SW-1&FIX_RES_DL_INTRF_RANDOM_SCH_SW-1;
ADD NRDUCELLRES: NrDuCellId=7, CellResId=1, ResUsageObject=NETWORK_SLICE_GROUP,
ResStrategyType=TTI_ONLY, DlRbMaxRatio=30, DlRbAverageRatio=30, DlRbMinRatio=20, UlRbMaxRatio=30,
UlRbAverageRatio=30, UlRbMinRatio=20;
//Configuring network slice groups for the NR DU cell and binding the NR DU cell resources to the groups
ADD NRDUCELLNSGRP: NrDuCellId=7, NsGrpId=0, CellResId=0, NsGrpDescription="GRP_DU";
ADD NRDUCELLNSGRP: NrDuCellId=7, NsGrpId=1, CellResId=1, NsGrpDescription="GRP_DU";
//(Required only for URLLC network slices) Configuring the network slices in the network slice groups of the
NR DU cell
ADD NRDUCELLNSGRPNS: NrDuCellId=7, NsGrpId=0, OperatorId=0, SliceServiceType=2,
SliceDifferentiator=6543;
ADD NRDUCELLNSGRPNS: NrDuCellId=7, NsGrpId=1, OperatorId=1, SliceServiceType=2,
SliceDifferentiator=6543;
//Configuring the uplink scheduling priority policy for UEs in a network slice group in an NR DU cell
MOD NRDUCELLULSCH: NrDuCellId=7, UlRbResSchPriority=NOT_CONFIG;
//Turning on the switch for the single-UE multi-slice RB management policy only when the minimum
downlink RB start position is set to 65535
MOD NRDUCELLALGOSWITCH: NrDuCellId=7,
NetworkSliceAlgoSwitch=SINGLEUE_MULTISLICE_RB_CTRL_SW-1;
//Configuration for the interfering neighboring cell, which is required if uplink/downlink interference
randomization has been enabled
//Enabling network slicing enhancement
MOD NRCELLALGOSWITCH: NrCellId=9, NetworkSliceAlgoSwitch=ENHANCED_NETWORKSLICE_SW-1;
//Enabling dynamic RB control
MOD NRDUCELLALGOSWITCH: NrDuCellId=9, NetworkSliceAlgoSwitch=RB_DYNAMIC_CONTROL_SW-1;
//Configuring NR DU cell resources and the RB resource control policy
ADD NRDUCELLRES: NrDuCellId=9, CellResId=0, ResUsageObject=NETWORK_SLICE_GROUP,
ResStrategyType=TTI_ONLY, DlRbMaxRatio=30, DlRbAverageRatio=30, DlRbMinRatio=30, UlRbMaxRatio=30,
UlRbAverageRatio=30, UlRbMinRatio=30, DlMinRbStartPos=0, UlMinRbStartPos=0,
SchedulingPolicySwitch=FIX_RES_UL_INTRF_RANDOM_SCH_SW-1&FIX_RES_DL_INTRF_RANDOM_SCH_SW-1;
//Configuring a network slice group for the NR DU cell and binding the NR DU cell resources to the group
ADD NRDUCELLNSGRP: NrDuCellId=9, NsGrpId=0, CellResId=0, NsGrpDescription="GRP_DU";
//(Required only for URLLC network slices) Configuring a network slice in the network slice group of the NR
DU cell
ADD NRDUCELLNSGRPNS: NrDuCellId=9, NsGrpId=0, OperatorId=0, SliceServiceType=2,
SliceDifferentiator=6543;
```

Configuring the Shared Resources Between the Network Slice Group and RFSP Group in an NR DU Cell (For details about RFSP groups, see *Resource Control in NSA Networking*.)

```
//Enabling dynamic RB control
MOD NRDUCELLALGOSWITCH: NrDuCellId=7, NetworkSliceAlgoSwitch=RB_DYNAMIC_CONTROL_SW-1;
MOD NRDUCELLCOMMONSW: NrDuCellId=7, RfspAlgoSwitch=RB_DYNAMIC_CONTROL_SW-1;
//Configuring NR DU cell resources and enabling intra-operator preferential sharing of idle resources and resource sharing based on operator-specific shares
ADD NRDUCELLRES: NrDuCellId=7, CellResId=2, ResUsageObject=NS_RFSP_GROUP, DlRbMaxRatio=10, DlRbMinRatio=10, DlRbAverageRatio=10, UlRbMaxRatio=10, UlRbMinRatio=10, UlRbAverageRatio=10;
//Configuring a network slice group and an RFSP group for the NR DU cell and binding the NR DU cell resources to the groups
ADD NRDUCELLNSGRP: NrDuCellId=7, NsGrpId=2, CellResId=2, NsGrpDescription="GRP_DU";
ADD NRDUCELLOPRFSP: NrDuCellId=7, OperatorId=1, RfspIndex=1, RfspGrpCellResId=2;
//Configuring a network slice in the network slice group of the NR DU cell
ADD NRDUCELLNSGRPNS: NrDuCellId=7, NsGrpId=2, OperatorId=1, SliceServiceType=1, SliceDifferentiator=0;
```

(Optional) Enabling Operator-specific Resource Sharing Enhancement After NR DU Cell Resources for Network Slices Are Configured or Resource Sharing Between Network Slice Groups and RFSP Groups Is Enabled for an NR DU Cell

```
MOD NRDUCELLNSMGMTCONFIG: NrDuCellId=7, NetSliceRfspGrpSharePol=SAME_OPERATOR_PREFERRED, NetSliceRfspGrpPreemptPol=BASED_ON_OP_SHARE;
```

Enabling User-Rate-based Dynamic Network Slice Resource Management

```
//Configuring the ID of the MAC parameter group referenced by a bearer in an NR DU cell, with the QCI specified for UEs in a network slice group (example uplink/downlink target rate: 10000)
MOD NRDUCELLQCIBEARER: NrDuCellId=7, Qci=128, MacParamGroupId=9, DynResUlTargetRate=10000, DynResDlTargetRate=10000;
//Turning on the switch for user-rate-based dynamic network slice resource management
MOD NRDUCELLCOMMONSW: NrDuCellId=7, ServiceDiffAlgoExtSwitch=DYNAMIC_NETWORKSLICE_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

NOTICE

The following deactivation command examples are provided based on the assumption of a single operator and a single network slice. In single-operator multi-network-slice or multi-operator multi-network-slice scenarios, all NR-cell network slice groups, NR-DU-cell network slice groups, and gNodeB-level network slices of each operator must be removed. Note that, in multi-operator multi-network-slice scenarios, operator-specific resource sharing enhancement must be deactivated.

Disabling Operator-specific Resource Sharing Enhancement (required only in multi-operator sharing scenarios)

```
MOD NRDUCELLNSMGMTCONFIG: NrDuCellId=7, NetSliceRfspGrpSharePol=FAIR_SHARING, NetSliceRfspGrpPreemptPol=BASED_ON_OP_SHARE;
```

Disabling User-Rate-based Dynamic Network Slice Resource Management

```
MOD NRDUCELLCOMMONSW: NrDuCellId=7, ServiceDiffAlgoExtSwitch=DYNAMIC_NETWORKSLICE_SW-0;
```

Disabling Network Slicing Enhancement

```
MOD NRCELLALGOSWITCH: NrCellId=7, NetworkSliceAlgoSwitch=ENHANCED_NETWORKSLICE_SW-0;
```

Removing a Network Slice Group from an NR Cell

```
//Removing a network slice from a network slice group of an NR cell
RMV NRCELLNSGRPNS: NrCellId=7, NsGrpId=0, OperatorId=0, SliceServiceType=1, SliceDifferentiator=6543;
//Removing the network slice group from the NR cell
RMV NRCELLNSGRP: NrCellId=7, NsGrpId=0;
//Removing NR cell resources
RMV NRCELLRES: NrCellId=7, CellResId=0;
```

Removing NR Cell Resources Shared Between the Network Slice Group and RFSP Group

```
//Removing the network slice from the network slice group of the NR cell
RMV NRCELLNSGRPNS: NrCellId=7, NsGrpId=0, OperatorId=0, SliceServiceType=1, SliceDifferentiator=6543;
//Removing the network slice group and RFSP group from the NR cell
RMV NRCELLNSGRP: NrCellId=7, NsGrpId=0;
RMV NRDUCELLOPRFSP: NrDuCellId=7, OperatorId=1, RfspIndex=1;
//Removing NR cell resources
RMV NRCELLRES: NrCellId=7, CellResId=0;
```

Removing the Network Slice Group from an NR DU Cell

```
//Disabling dynamic RB control
MOD NRDUCELLALGOSWITCH: NrDuCellId=7, NetworkSliceAlgoSwitch=RB_DYNAMIC_CONTROL_SW-0;
//Removing the network slice from the network slice group of the NR DU cell
RMV NRDUCELLNSGRPNS: NrDuCellId=7, NsGrpId=0, OperatorId=0, SliceServiceType=1,
SliceDifferentiator=6543;
//Removing the network slice group from the NR DU cell
RMV NRDUCELLNSGRP: NrDuCellId=7, NsGrpId=0;
//Removing NR DU cell resources
RMV NRDUCELLRES: NrDuCellId=7, CellResId=0;
```

Removing NR DU Cell Resources Shared Between the Network Slice Group and RFSP Group

```
//Disabling dynamic RB control
MOD NRDUCELLALGOSWITCH: NrDuCellId=7, NetworkSliceAlgoSwitch=RB_DYNAMIC_CONTROL_SW-0;
MOD NRDUCELLCOMMONSW: NrDuCellId=7, RfspAlgoSwitch=RB_DYNAMIC_CONTROL_SW-0;
//Removing the network slice from the network slice group of the NR DU cell
RMV NRDUCELLNSGRPNS: NrDuCellId=7, NsGrpId=0, OperatorId=0, SliceServiceType=1,
SliceDifferentiator=6543;
//Removing the network slice group and RFSP group from the NR DU cell
RMV NRDUCELLNSGRP: NrDuCellId=7, NsGrpId=0;
RMV NRDUCELLOPRFSP: NrDuCellId=7, OperatorId=0, RfspIndex=1;
//Removing NR DU cell resources
RMV NRDUCELLRES: NrDuCellId=7, CellResId=0;
```

Removing the Network Slice from the gNodeB

```
//Unbinding the network slice from the gNodeB CU NG object
RMV GNBCUNGNS: gNBCuNgId=0, OperatorId=0, SliceServiceType=1, SliceDifferentiator=6543;
//Unbinding the network slice from the tracking area ID
RMV GNBTANS: TrackingAreaId=0, OperatorId=0, SliceServiceType=1, SliceDifferentiator=6543;
//Removing the network slice from the gNodeB.
RMV GNBNETWORKSLICE: OperatorId=0, SliceServiceType=1, SliceDifferentiator=6543;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Using Counters

Check the values of the following performance counters to determine whether the network slicing function has been enabled. If the value of any of the counters is not 0, this function has been enabled.

- **N.PDUSession.Est.Succ.Ns**
- **N.PDUSession.Est.Att.Ns**
- **N.User.RRCConn.Avg.Ns**
- **N.ThpVol.UL.Ns**
- **N.ThpVol.DL.Ns**
- **N.QosFlow.Avg.Ns**
- **N.QosFlow.Max.Ns**
- **N.QosFlow.AMF.Mod.Succ.Ns**
- **N.QosFlow.AMF.Mod.Att.Ns**
- **N.QosFlow.Est.Succ.Ns**
- **N.QosFlow.Est.Att.Ns**
- **N.QosFlow.AbnormRel.Ns**
- **N.QosFlow.NormRel.Ns**
- **N.QosFlow.AMF.Mod.Att.Ns.EPSFB**
- **N.QosFlow.Est.Att.Ns.EPSFB**
- **N.QosFlow.InitEst.Succ.Ns**
- **N.QosFlow.InitEst.Att.Ns**
- **N.QosFlow.AbnormRel.AMF.Ns**
- **N.QosFlow.NormRel.AMF.Ns**
- **N.QosFlow.FailEst.TNL.Ns**
- **N.QosFlow.AbnormRel.TNL.Ns**
- **N.PDCP.Vol.UL.TrfPDU.Rx.Ns**
- **N.PDCP.Vol.DL.TrfPDU.Tx.Ns**
- **N.QoS.DL.PktDelaygNBDU.Num.Ns**
- **N.QoS.DL.PktDelaygNBDU.Time.Ns**
- **N.QoS.DL.PktDelayAirInterface.Time.Ns**
- **N.QoS.DL.PktDelayAirInterface.Num.Ns**
- **N.ThpVol.DL.LastSlot.Ns**
- **N.ThpTime.DL.RmvLastSlot.Ns**
- **N.PRBUsed.UL.NsGrp0.Avg** to **N.PRBUsed.UL.NsGrp31.Avg**
- **N.PRBUsed.DL.NsGrp0.Avg** to **N.PRBUsed.DL.NsGrp31.Avg**
- **N.ThpVolForRate.UL.Ns**
- **N.ThpVol.UE.UL.SmallPkt.Ns**
- **N.ThpTime.UE.UL.RmvSmallPkt.Ns**

- **N.PRBUsed.DL.Avg.Ns**
- **N.HO.IntraRAT.PrepAttOut.Ns**
- **N.HO.IntraRAT.ExecAttOut.Ns**
- **N.HO.IntraRAT.ExecSuccOut.Ns**
- **N.PRBUsed.UL.Avg.Ns**
- **N.User.RRCConn.Max.Ns**
- **N.User.RRCConn.Active.Max.Ns**
- **N.User.RRCConn.Active.Avg.Ns**
- **N.BWP.ThpTime.DL.RmvLastSlot.Ns**
- **N.QosFlow.HoOut.Ns**
- **N.QosFlow.Est.Att.EmcFB.Ns**
- **N.BWP.ThpVol.DL.LastSlot.Ns**
- **N.BWP.ThpVol.DL.Ns**
- **N.QosFlow.FailEst.AMF.SyntaxError.Ns**
- **N.PDUSession.FailEst.AMF.SyntaxError.Ns**
- **N.QosFlow.RrcConnToInactive.Suspend.Ns**
- **N.QosFlow.RrcInactiveToIdle.Rel.gNB.Ns**
- **N.HO.IntraFreq.IntraRAT.PrepAttOut.Ns**
- **N.HO.IntraFreq.IntraRAT.ExecSuccOut.Ns**
- **N.PDCP.UL.TrfPDU.RxPackets.Ns**
- **N.PDCP.UL.TrfSDU.RxPacket.Loss.Ns**
- **N.PDCP.DL.TrfPDU.TxPackets.Ns**
- **N.PDCP.DL.TrfSDU.TxPacket.Discard.Ns**
- **N.RLC.DL.TrfSDU.PktUuTrans.Loss.Ns**
- **N.RLC.DL.TrfSDU.RxPackets.Ns**
- **N.RLC.DL.TrfSDU.RxPackets.DuCell5QI.Ns**
- **N.QosFlow.FailEst.ResNotAvailForNs.Ns**
- **N.QosFlow.FailEst.UeNoReply.Ns**
- **N.QosFlow.FailEst.NoRadioRes.Ns**

Using MML Commands

Run the **DSP GNBNETWORKSLICE** command. If **Slice Service Initiation Count** has a value other than 0 in the command output, the network slicing function has been successfully activated.

4.4.3 Network Monitoring

After the network slicing function is activated, observe performance counters to monitor the network slice services in a cell. For details about the counters, see 4.4.2 Activation Verification. Alternatively, run the **DSP GNBNETWORKSLICE** command to query the configuration and status of a network slice.

5 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 38.300: "NR; NR and NG-RAN Overall Description; Stage 2"
- 3GPP TS 23.501: "System architecture for the 5G System (5GS); Stage 2"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.214: "NR; Physical layer procedures for data"
- 3GPP TS 38.413: "NG-RAN; NG Application Protocol (NGAP)"
- *DRX*
- *Performance Management*
- *Adaptive Managed Latency*
- *MIMO (TDD)*
- *NG and Xn Self-Management*
- *CoMP*
- *Coordinated Interference Management (Low-Frequency TDD)*
- *Co-MM (Low-Frequency TDD)*
- *Carrier Aggregation*
- *UL and DL Decoupling*
- *Multi-Operator Sharing*
- *URLLC*
- *Cell Combination*
- *FWA*
- *Energy Conservation and Emission Reduction*
- *VoNR*
- *UE Power Saving*
- *Uplink Scheduling*
- *Downlink Scheduling*
- *Super Uplink*
- *SA Mobility Management in Connected Mode*
- *Admission Control*
- *Mobility Load Balancing in Connected Mode*

- *QoS Management*
- *Carrier Aggregation*
- *RedCap*
- *License Control Item Lists*
- *Resource Control in NSA Networking*
- *Cell Management*
- *QCI-based Resource Control*
- *Service-differentiated Experience Guarantee*
- *Massive MIMO iBeam (Low-Frequency TDD)*
- *High Speed Mobility*
- *Hyper Cell*
- *Technical Specifications* in *3900 & 5900 Series Base Station Product Documentation*